

The Alberta Buck - Ethereum Implementation (DRAFT v2)

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Abstract

A concrete Ethereum implementation of the Alberta Buck Architecture, in six contracts:

1. **BUCK_CREDIT** (ERC-721) – one NFT per insured asset, with deterministic depreciation and piecemeal client activation.
2. **IdentityRegistry** – per-address Pointcheval-Sanders credentials and ElGamal-encrypted identity points, verified on-chain via the BN254 precompiles. Trust anchor for every BUCK transfer.
3. **BUCK** (ERC-20) – identity-bound, single-slot per-account state with *signed* raw balances, continuous demurrage on positive holdings, and NFT-backed credit headroom that lets the raw balance go negative up to $\text{creditLimit}(a) = \text{totalCurrentValue}(a) * \text{currentBuckK} / 1e18$. $\text{mint}(N)$ activates NFT-backed coverage and pays the insurance pool principal out of the holder's credit (no BUCK is delivered to the holder's raw balance); the holder spends the new headroom by transferring out, driving the signed raw negative. balanceOf reports $\text{held} + \text{unused credit}$ on non-Carrying accounts, so the ERC-20 view answers "*what can I spend right now*".
4. **BuckBasket** – USD-free basket registry and direct-mint orchestrator. Custodies a single full-range Uniswap-V3 LP position per TOKEN/BUCK pool, mints BUCK against deposited basket tokens at the current pool spot price, and on redemption withdraws proportionally from overweight pools, hands the depositor the TOKEN side(s) and recycles the BUCK side "sell high / buy low" into the most underweight pool as treasury LP. Publishes $\text{basketValueInBuck}() = \sum(\text{basketAmount}_i * \text{pool_price_in_BUCK}_i)$ as the direct-embodiment process variable for BUCK_K.
5. **BUCK_K** – the value-stabilization factor for credit-limit computation. Three implementations ship today: **BuckKControllerStatic** (governance-set constant), **BuckKControllerDirect** (USD-free PID; setpoint = 1.0 BUCK by definition, process variable = $\text{BuckBasket.basketValueInBuck}()$), and **BuckKController** (PID against an external four-pool USDT/USDC basket of Uniswap-V3 TWAPs, with a BUCK/USDT TWAP as the BUCK price). The PID mechanics

– priming, setpoint-shift (dS) compensation, long-gap dTMax clamp, `fundingFactor()` counter-cyclical insurance view – live in a shared `BuckKControllerBase` following the `ezpwd::pid` pattern; `Buck.mint` and `Buck.burn` drive `compute()`, and `BuckBasket` holds the privileged hook to `reprime()` after constituent additions.

6. **Notes** – privacy-preserving denomination on top of BUCK. Mint and spend via Groth16 SNARKs over a Tornado-style incremental Poseidon Merkle tree.

This is the first implementation, prioritizing clarity over breadth of standards. Subsequent documents will examine ERC-1155, ERC-3643, and ERC-4626 alternatives for individual layers.

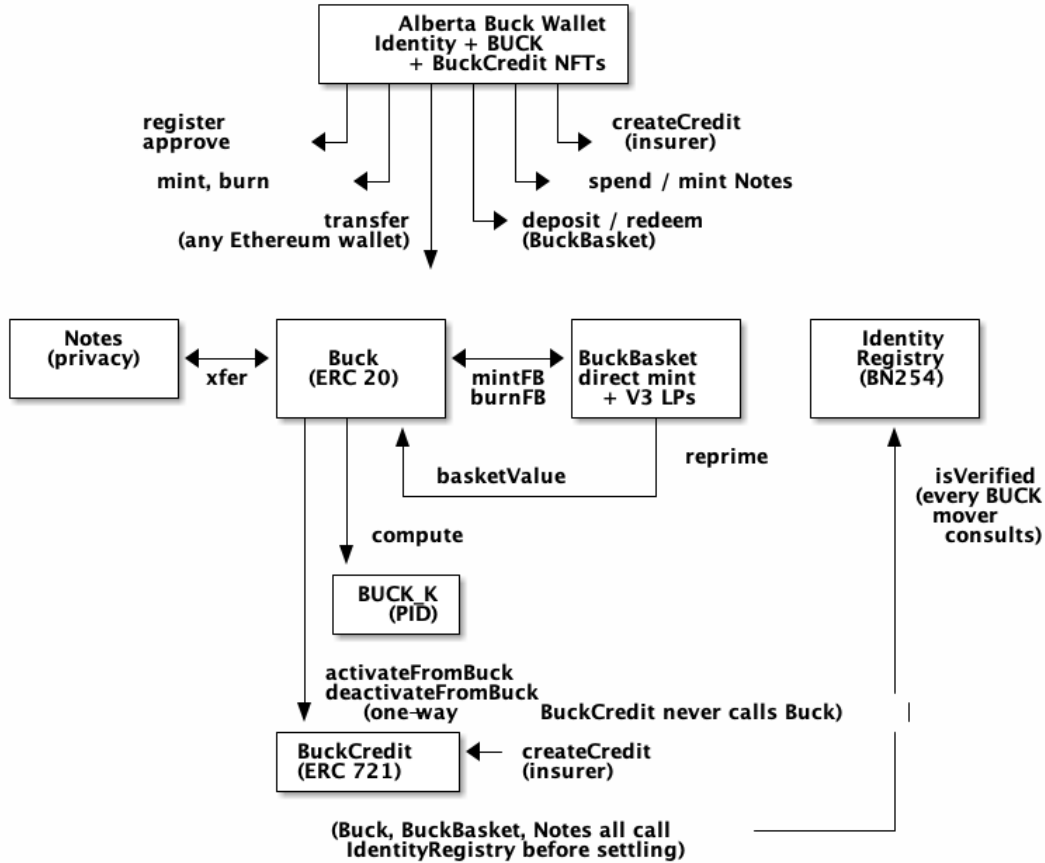
Contents

1	Contract Architecture	5
2	ERC Standard Choice for BUCK_CREDIT	7
3	BUCK_CREDIT: ERC-721 Insured Asset NFT	7
3.1	Data Model	7
3.2	Depreciation Models	8
3.3	Marked Credits	8
3.4	Activation = Mint, Deactivation = Burn	9
3.5	Buck Wiring	10
3.6	Insurer Updates	10
4	IdentityRegistry: On-Chain Trust Anchor	10
4.1	Cryptographic Surface	11
4.2	Solidity Surface	11
4.3	Public-Identity Contracts	13
4.4	Issuer Trust	13
4.5	The Identity Accumulator	13
5	BUCK: ERC-20 Token	14
5.1	Storage Layout	14
5.2	Signed Balances and <code>balanceOf</code>	15
5.2.1	The Credit Limit Is Read Live	16
5.3	<code>mint(N)</code> and <code>burn(N)</code>	16
5.3.1	The Per-Credit Inversion	17
5.3.2	Releasing: the Deposit Comes Back	17
5.3.3	The Funding-Factor Reserve, and Its Precedents	20
5.3.4	Underwater Is a State, Not a Failure	21
5.4	Identity-Bound <code>approve()</code>	21
5.5	<code>transfer()</code> and <code>transferFrom()</code>	22
5.5.1	Reentrancy Guard	23
5.6	Demurrage and Jubilee	23
5.6.1	Delegated Demurrage: a Designated Fee Payer	25
5.7	Integrating with BUCK	25
6	BuckBasket: Direct-Mint Orchestrator + USD-Free Value Anchor	26
6.1	Two Kinds of BUCK	26
6.2	Constituents and Weights	27
6.3	Process Variable: <code>basketValueInBuck()</code>	27
6.4	Direct-Mint: <code>depositToken</code>	28
6.5	Redemption: "Sell High, Recycle to Buy Low"	28
6.6	Invariant	29
6.7	Known Gaps	29
6.8	Dynamic Issuance: Why BuckBasket Needs BUCK_K	29

7	BuckBasketEquity: the Basket as a Credit Holder	31
7.1	The Account and the Credit	31
7.2	Valuation	31
7.3	Deposit	31
7.4	Redemption	32
7.5	The Wheel's Components	32
7.6	The Director	32
7.7	The Desk: Its Own Credit Holder	33
7.8	Invariants	33
8	BUCK_K: Value Stabilization Controller	34
8.1	The Loop	34
8.2	Three Controllers	34
9	Notes: Privacy-Preserving Denomination	35
9.1	Mint Circuit (Batch, In-Circuit Merkle Insertion)	35
9.2	The Spend Proof and the Deposit Gates	36
9.3	Verifier Interfaces	37
9.4	Notes Contract	37
10	End-to-End Walkthrough	38
10.1	Step 0: Identity Registration (One-Time)	38
10.2	Step 1: Mint	39
10.3	Step 2: Approve Bob (Identity Handshake)	39
10.4	Step 3: Bob Pulls the Transfer	40
10.5	Step 4: Alice Denominates as Notes	40
10.6	Step 5: Spend a Note	40
11	Gas Cost Snapshot	41
12	Security Considerations	41
12.1	Oracle Manipulation	41
12.2	The Insurer Gate	42
12.3	Recipient Opt-In	43
12.4	Insurer Trust	43
12.5	Reentrancy	43
12.6	The Insurance Pool: an Interim Stand-in	44
13	Future Directions	45
14	Status	46

1 Contract Architecture

The six contracts compose into a single transfer pipeline gated on identity. `IdentityRegistry` is the trust anchor: `mint`, `approve` and `transfer` all consult it before mutating state. Every account *and* contract that touches BUCK carries an identity binding – EOAs via `register`, contracts via `bindContract` – so bilateral identity-checked transfers always have a verifiable counterparty on both sides.



A Wallet (EOA) is the only initiator. It registers once with `IdentityRegistry` and then mints, burns, transfers, deposits or spends via the contract entry points in the second row. Every contract that moves BUCK – Buck, BuckBasket, Notes – consults `IdentityRegistry.isVerified` on both counterparties before settling. `BuckCredit` (ERC-721) is mutated only by insurers (`createCredit`, `updateCredit`) and by Buck itself (`activateFromBuck` / `deactivateFromBuck` during mint and burn); the dependency is one-way – `BuckCredit` never calls back into Buck. `BUCK_K` is private infrastructure: Buck calls `compute()` on every mint/burn, and `BuckBasket` is the privileged caller of `reprime()` and the source of `basketValueInBuck()` that the Direct variant integrates against. External dependencies (not shown): Uniswap V3 (LPs custodied by `BuckBasket`; TWAPs read by the External variant of `BUCK_K`); the BN254 EVM precompiles (pairings for `IdentityRegistry`, modular arithmetic for the `Mint/SpendVerifier` contracts); circom / snarkjs for off-chain Notes proof generation.

The flow is:

1. A **client** names an insurer it will accept credits from (**setCreditIssuer**), and that **insurer** mints a BUCK_CREDIT NFT for the client's asset, setting face value, depreciation schedule, and **premiumRate** (annual basis points of activated coverage). A credit only lands where its recipient asked for it, and only from an insurer whose regulator has attested it – see The Insurer Gate.
2. There is no separate activation step, and no public **BuckCredit.activate**. Coverage is activated only inside **Buck.mint**, atomically with the insurance principal that funds it – see Activation = Mint, Deactivation = Burn. While coverage is live the credit cannot change hands.
3. Before touching BUCK, the client **registers** with **IdentityRegistry**, posting a blinded presentation (A, B) of its PS credential, an ElGamal-encrypted identity point and a proof tying the two to the account key. The registry verifies the proof via BN254 pairings (443,892 gas measured, paid once per address).
4. **BUCK.mint(amount[, tokenIds])** draws coverage cheapest-first across the client's NFTs, activates it, and debits the mutual-insurance pool principal – 10x the annual premium on the coverage – from the client's *signed* raw balance into **insurancePool1**. No BUCK is delivered to the client's raw balance; what the client gains is spendable headroom. The arithmetic is in **mint(N)** and **burn(N)**.
5. To send BUCK, the sender **approves** with a Chaum-Pedersen receipt that re-encrypts the sender's identity under the spender's public key. **transfer** / **transferFrom** then bilateral-identity-check both sides and emit **BuckTransferReceipt** carrying only ciphertext hashes.
6. To denominate as private notes, the client calls **Notes.mint** with a Groth16 proof binding the deposited **totalFace** to a batch of Poseidon commitments. Holders later call **Notes.spendCoupledA1** / **A2** / **B1** with a fresh nullifier, a SNARK proving inclusion under a recent root, and a deposit gate proving the depositor is the identity the note names, against a posted identity root.
7. **BuckBasket** is the **direct-mint orchestrator** and the value anchor. Anyone holding a basket constituent (PAXG, cbBTC, XAUT, WBTC, etc., as wired by governance via **addBasketToken**) can call **BuckBasket.depositToken** to LP it into the corresponding TOKEN/BUCK Uniswap-V3 pool; the basket mints exactly **tokenAmount * spotPrice** worth of fresh BUCK against the deposit (via the privileged **Buck.mintFromBasket** hook), pairs the two sides as full-range V3 liquidity, and hands the depositor an ERC-721 receipt (**BuckBasketReceipt**). Depositing already-minted BUCK is symmetric: the basket swaps it into the most underweight constituent's TOKEN and LPs there. On **BuckBasket.redeem**, the value claim is allocated across overweight pools proportional to each pool's positive value-weight error; the basket withdraws L from each, transfers the entire TOKEN side to the depositor, burns the principal BUCK from the BUCK side, and routes the remaining "profit" BUCK into the most underweight pool's TOKEN as treasury LP – the "sell high on the way out, recycle to buy low" leg fires in the same transaction.
8. BUCK_K ships in three forms today. The governance-set static variant is the simpler choice for early deployments where no on-chain price reference exists. **BuckKControllerDirect** is the **USD-free** embodiment: process variable = **BuckBasket.basketValueInBuck()**, setpoint = 1.0 BUCK by definition, error = 1.0 - **basketValue**. The legacy **BuckKController** reads its process variable (BUCK price) from a BUCK/USDT TWAP and its setpoint (basket

cost) from a four-pool USDT/USDC basket – XAUT/USDT, PAXG/USDC, cbBTC/USDC, WBTC/USDT, 25 % each – chosen for resilience against any single basis-currency or asset-token freeze. `Buck.mint` and `Buck.burn` call `BuckK.compute()` regardless of variant, so user activity drives and amortizes the PID; long quiet stretches are bounded by `dTMax`, and `BuckBasket` is the privileged caller of `reprime()` after `addBasketToken` so dilution discontinuities don’t fire a one-cycle P/I spike.

2 ERC Standard Choice for BUCK_CREDIT

ERC-721 was the obvious starting point and the choice that ships. Each credit is genuinely unique (specific asset, depreciation curve, insured value, premium schedule); `ERC721Enumerable` gives `BUCK.mint` the `tokenOfOwnerByIndex` it needs to walk a holder’s NFTs. ERC-3643 (T-REX) is a plausible future move if regulatory KYC becomes a hard requirement; ERC-1155 helps only if standardized credit *classes* emerge with shared depreciation parameters; ERC-4626 fits the `InsurancePool` but not individual credits. None of those are revisited below – the rest of the document specifies the shipped ERC-721 path.

3 BUCK_CREDIT: ERC-721 Insured Asset NFT

Each `BUCK_CREDIT` is an `ERC721Enumerable` token whose per-token `CreditParams` holds the insurer’s current offer and the client’s activation state. Three roles touch it: the **insurer** mints and updates, the **client** activates, `BUCK` reads the depreciated activated value to size mint allocations. All monetary fields are in 6-decimal `BUCK`-equivalent units (USDC-compatible).

3.1 Data Model

Fields group by mutability:

Issuance is opt-in per (client, insurer) pair: `createCredit` reverts unless the client has called `setCreditIssuer(insurer, true)`, and – where the gate is configured – unless the insurer’s regulator has attested an envelope covering the credit (The Insurer Gate). Nobody can put a credit in your hands uninvited, or write one beyond their licence.

- **Immutable at mint:** `insurer` (the address allowed to update this token), `assetClass`, `createdAt`.
- **Insurer-mutable** via `updateCredit`: `faceValue`, `depreciationFloor`, `depType`, `depRate`, `depStartAt`, `premiumRate`, `lastUpdated`. `faceValue` may not be reappraised below `activatedValue` – see Insurer Updates.
- **Buck-mutable** via `activateFromBuck` / `deactivateFromBuck`: `activatedValue` (capped at `faceValue`). The client never writes it directly.

`DepreciationType` is one of `NONE` (land, gold), `LINEAR` (constant annual reduction), `DECLINING_BALANCE` (continuous-compounded percentage), or `MARKED` (valued at its insurer’s latest mark: Marked Credits). Four events – `CreditCreated`, `CreditUpdated`, `CreditActivated`, `CreditMarked` – record every state change.

A credit does not age. Jubilee relief accrues on the `BUCK` an account has actually drawn – its lien, in `Buck` – not on coverage (Demurrage and Jubilee).

3.2 Depreciation Models

`currentValue(tokenId)` is a pure view that branches on `depType`:

- **NONE**: returns the activated face value, unchanged.
- **LINEAR**: loss accrues at `depRate` bp/year on the depreciable portion (`faceValue - floor`), clamped at `depreciationFloor`.
- **DECLINING_BALANCE**: discrete whole-year compounding – the factor $(BP - \text{depRate})/BP$ is applied once per complete year via a loop capped at 100 years, then a linear interpolation bridges the trailing partial year. No transcendental approximations.

In every case the activated portion depreciates proportionally: `currentValue = depreciatedFace * activatedValue / faceValue`.

Asset	Type	Rate	Floor	30yr value (of \$100K)
Land	NONE	–	–	\$100,000
House (bldg)	LINEAR	200bp	20%	\$52,000
Vehicle	DECLINING_BALANCE	1500bp	5%	\$5,725
Farm equip	DECLINING_BALANCE	1000bp	10%	\$13,815
Gold in vault	NONE	–	–	\$100,000

A residential property would typically have two BUCK_CREDITS: one for the land (NONE) and one for the structure (LINEAR or DECLINING_BALANCE with a salvage floor).

3.3 Marked Credits

Some collateral is revalued on-chain every block – a BuckBasket’s equity is the case in point – and no depreciation schedule describes it. A MARKED credit is valued at a mark its insurer keeps current, `mark(tokenId, value)`:

```
depreciatedFaceValue(tokenId) = min(faceValue, markOf[tokenId])      (MARKED)
currentValue(tokenId)          = depreciatedFaceValue * activatedValue / faceValue
```

so with the whole face activated, `currentValue` is the mark and the holder’s `creditLimit` is $K \times \text{mark}$. Like depreciation, the mark moves the limit without touching the coverage: it may fall below `activatedValue`, and a holder whose lien then exceeds its limit is under water – it cannot spend further credit, and nothing is called back.

That is exactly what an insurer must never be able to do to someone else: a mark written down on another’s credit would revoke a purchased policy, which Insurer Updates forbids. So three rules bound it:

- **Self-issued only.** `createCredit` with MARKED requires `client = msg.sender`: the holder is its own insurer, and marks only its own collateral.
- **Fixed at creation.** `updateCredit` refuses to switch a credit to or from MARKED, so a scheduled credit cannot be converted and then marked down.
- **Gated like any model.** Where the insurer gate is configured, the insurer’s attested envelope must permit MARKED (bit 3 of `depTypes`), as it must permit any depreciation model.

A marked credit is activated like any other – one `Buck.mint` at zero premium, once the mark is above zero (a credit appraised at zero insures nothing and is skipped) – and its holder’s lien earns relief like any other. Its face caps what the holder can ever draw, at $K \times \text{face}$.

3.4 Activation = Mint, Deactivation = Burn

There is no public `BuckCredit.activate()` function. Activation – expanding `activatedValue` and thus growing the holder’s `creditLimit` – happens *only* atomically as part of `Buck.mint(N, [tids])`, and deactivation happens *only* atomically as part of `Buck.burn(N, [tids])`.

This is a deliberate consequence of the mutual-insurance pool model. Drawing `take` units of coverage on an NFT generates `annual_premium = take * premiumRate / BP` per year. Instead of charging that premium on a continuous schedule, the contract requires an *upfront* `poolPrincipal = annual_premium * POOL_ROI_INV` deposit (where `POOL_ROI_INV = 10` encodes the insurer’s assumed 10 % annual ROI on pool capital). At parity, the pool’s investment yield on `poolPrincipal` exactly funds the annual premium on `take` in perpetuity – so *a policy is a one-time purchase*, not a recurring expense.

Therefore the surface `BuckCredit` exposes is restricted to (a) insurer-side creation / re-appraisal (`createCredit`, `updateCredit`), (b) standard ERC-721 ownership operations (`transferFrom`, `balanceOf`, `ownerOf`, ...), and (c) two Buck-only entry points (`activateFromBuck`, `deactivateFromBuck`) which `Buck._allocateMint` / `_allocateBurn` call as part of their atomic sequences. Test fixtures that need to set up a holder with non-zero `creditLimit` without exercising the `mint()` purchase path use the `BuckCreditHarness` subclass’s `forceActivate` – explicitly not present on production `BuckCredit`.

`Buck.mint` and `Buck.burn` are the only callers. Their per-NFT arithmetic, and the post-burn solvency rule that goes with it, are in `mint(N)` and `burn(N)`.

A credit that is backing BUCK cannot change hands. `BuckCredit._update` – the ERC-721 hook every mint, transfer and burn passes through – refuses to move a token whose `activatedValue` is non-zero, with `BuckCredit: credit in use`. Minting is exempt (`from = address(0)=`), so `createCredit` still works, and a burn that takes coverage back to zero frees the credit to be sold again.

The reason is that the two halves of a credit position live in different places. `Buck` derives the holder’s `creditLimit` from the activated value of the credits they *own*, while the BUCK drawn against that limit stays as a negative raw balance on the account that *drew* it. Let the token move and the two separate: the seller keeps an obligation with nothing behind it, and the buyer receives headroom against coverage that has already been spent. The same activated value would back BUCK twice.

`activatedValue` is the predicate rather than `Buck`’s `mintsBacked` deliberately: it is local, so `BuckCredit` enforces the property without consulting `Buck` at all, and the guarantee does not depend on another contract being correct or even reachable. The two agree by construction – activation happens only inside `Buck._allocateMint`, and `updateCredit` may not reappraise below activated coverage.

`totalCurrentValue(account)` walks `tokenOfOwnerByIndex` and sums `currentValue(tokenId)` – the single call `Buck.creditLimit` makes to size the holder’s headroom (then multiplied by `currentBuckK` and divided by `1e18`). `creditInfo(tokenId)` returns (`faceValue`, `activatedValue`, `premiumRate`) one NFT at a time; `batchCreditInfo(uint256[] tokenIds)` returns a packed `CreditSlice[] = (owner, faceValue, activatedValue, premiumRate)` for the entire list in one external call, used by `Buck._allocateMint` / `_allocateBurn` to amortize per-NFT cross-contract dispatch overhead.

3.5 Buck Wiring

The dependency is one-way: `Buck` reads `BuckCredit` to size credit, and `BuckCredit` never calls `Buck`. `BuckCredit.setBuck(address)` is a one-shot post-deployment authorisation record – it names the single contract allowed to call `activateFromBuck` / `deactivateFromBuck` – and nothing more.

Earlier drafts had `BuckCredit` notify `Buck` on every NFT mutation, so that a per-block credit-limit cache in `Buck` could invalidate. Both the cache and the notification are gone; see *The Credit Limit Is Read Live* for why the cache could not have been made correct. Removing them left the call graph acyclic, which is worth more than the cache was: there is no reentrant path from `BuckCredit` back into `Buck`, so `Buck`’s reentrancy guard needs no exemption for one.

The two-contract split, rather than collapsing into a single `Diamond`, is forced by the ERC-20 / ERC-721 standards: `balanceOf(address)` and `transferFrom(address,address,uint256)` share function selectors with incompatible semantics across the two standards, and the `Transfer` event has different indexed-argument counts. A `Diamond` can dispatch one selector per signature; whichever loses stops being standards-compliant and silently breaks wallets, indexers, marketplaces, and routers. See *Future Directions* for the modularization paths that *are* viable (intra-ERC-20 facets in `Buck`, independent UUPS for `BuckCredit`).

3.6 Insurer Updates

`updateCredit(tokenId, newFaceValue, newDepreciationFloor, newDepType, newDepRate, newDepStartAt, newPremiumRate)` is gated on `msg.sender == credits[tokenId].insurer.assetClass` and `insurer` never change.

An insurer may reappraise down to the coverage the holder has already bought, and no further: `newFaceValue >= activatedValue`, else `BuckCredit`: `face` below `activated coverage`. Activated coverage is a completed purchase – its pool principal was paid up front and funds its premium in perpetuity – so writing it down revokes a policy rather than revaluing an asset. An asset that has genuinely lost value is what a **claim** is for; ordinary decline is what the depreciation schedule is for, and that still moves the holder’s credit limit continuously without touching the coverage itself. Every other lever stays: `faceValue` down to the activated line, the schedule, the floor, the premium rate.

The rule is also load-bearing for solvency. `activatedValue` lives in `BuckCredit` and `mintsBacked` lives in `Buck`; the burn is sized from the second and checked against the first. An earlier draft let a reappraisal clamp one without the other, and a holder whose credit was written down could then never unwind the position – burnable only down to the clamped line, with the remainder stuck and the credit permanently disqualified from backing BUCK again. Refusing the clamp keeps the two equal by construction, which is also what makes the transfer lock in `Activation = Mint` the right predicate.

A reappraisal, or ordinary depreciation, can still leave a holder’s outstanding BUCK above their credit limit. That is a supported state, not a default: see *Underwater Is a State, Not a Failure*.

4 IdentityRegistry: On-Chain Trust Anchor

The `IdentityRegistry` is the binding between Ethereum addresses and the privacy-preserving identity machinery described in the companion documents *Identity with Anonymity*, *Identity Example* – *Data Flow*, and *Cryptographic Proofs*. Every BUCK `mint`, `approve` and `transfer` consults this registry; an unregistered address cannot hold or move BUCKs.

The registry stores per address: `pk[a]` (G₁ identity public key), `E_addr[a]` (ElGamal ciphertext of the identity point M), `isVerified` (true after PS+NIZK pass on-chain or after `bindContract`), `isCarrying` (controls Buck's transfer dispatch), and `isPublicIdentity` (true only for contracts bound with the public-disclosure flag). Trusted-issuer PS public keys live in `trustedIssuers[issuerAddr]`, populated by governance. EOAs always self-register; contracts always `bindContract`.

4.1 Cryptographic Surface

Four on-chain verifications, all reducible to BN254 precompile calls:

1. **PS credential presentation at registration:** the holder publishes $(A, B) = (a * \sigma_1, a * \sigma_2 + b * Y_1)$ for fresh a, b , with $Y_1 = y * G$ the issuer's G₁ key component, and the registry checks the masked relation $e(B, g_2) = e(A, X) * e(m * A + b * G, Y)$ through the proof in step 2 – one three-pair pairing product. The published pair is uniform whatever the identity, so no holder of m (the issuer included) can test it; a rerandomized signature, which an earlier design published, would be a valid signature on m and testable by anyone holding it.
2. **Schnorr-family NIZK** binding the presentation to the ElGamal ciphertext and the account key. Four response scalars over one credential commitment $C_1 = m * A + b * G$ (so no field exposes $m * A$ alone), Fiat-Shamir bound to the action/version domain (AlbertaBuck/FiatShamir/IdentityRegistry/Register/v2), $((A, B), E_{\text{new}}, pk_{\text{new}}, registrant_addr)$, chain ID, and registry deployment. Total registration 443,892 gas measured (472,630 before the presentation), paid once per address; `trustIssuer` additionally checks $e(Y_1, g_2) = e(G, Y)$ once per issuer.
3. **Chaum-Pedersen NIZK at approve:** proves the recipient's identity ciphertext re-encrypts the same M registered in `E_alice` and that the sender holds the registered account key. Fiat-Shamir binds the registry deployment, `msg.sender`, `spender`, `chainid` and the `.../Approve/v2` domain word. ~29K gas per counterparty pair.
4. **The Notes bindings**, each a sigma over EIP-196 with its own v2 domain word: the public issuer's batch Schnorr (`verifyIssuerSchnorr`), the private issuer's recipient-blinded re-encryption (`verifyIssuerReenc`), and the bearer depositor's binding (`verifyDepositorBinding`). The addressed deposit gate is a folded Groth16 proof, not a sigma (Notes).
5. **Public membership** (`verifyPublicMembership`): a Poseidon path through a public subtree and the aggregator, checked on chain for the insurer gate.

Every hash the protocol relies on carries a domain tag of the form `AlbertaBuck/<Area>/<Name>/v2`: the Fiat-Shamir transcripts end in one, the identity scalar hashes one ahead of the certified record, and each accumulator leaf leads with one as a field element.

4.2 Solidity Surface

`src/IdentityRegistry.sol` uses the BN254 helper for all curve math:

```
struct G1Point { uint256 X; uint256 Y; }
struct G2Point { uint256[2] X; uint256[2] Y; }
```

```

struct ElGamalCT { G1Point R; G1Point C; }      // (R, C) = (r*G, r*pk + M)
struct PSPresentation { G1Point A; G1Point B; }  // (a*sigma_1, a*sigma_2 + b*Y1)
struct PSPubKey { G2Point X; G2Point Y; G1Point Y1; }  // Y1 = y*G, checked at trustIssuer

struct RegistrationProof { uint256 e, s_m, s_b, s_r, s_sk; G1Point C1, T_C, T_R, T_key; }
struct CPProof { uint256 e, s1, s2; G1Point T1, T2, T3; }
struct ContractBindingProof { uint256 e, s; G1Point T; }
struct SchnorrProof, IssuerReencProof, DepositorBindingProof;  // the Notes bindings

function register(address issuer, G1Point pk, ElGamalCT E,
                  PSPresentation pres, RegistrationProof proof);
function authorizeContractBinding(address binder, G1Point pk, ElGamalCT E,
                                  bool isPublicIdentity_, bool isCarrying_);
function bindContract(address target, G1Point pk, ElGamalCT E,
                      bool isPublicIdentity_, bool isCarrying_);
function bindContract(address target, address issuer, G1Point pk, ElGamalCT E,
                      PSPresentation pres, RegistrationProof credential,
                      ContractBindingProof holderAuthorization,
                      bool isPublicIdentity_, bool isCarrying_);
function bindContractFromAdapter(address target, address operator,
                                  bool isPublicIdentity_, bool isCarrying_);
function setBindingAdapter(address adapter, bool approved);           // governance
function trustIssuer(address issuer, PSPubKey ipk);                  // governance
function verifyApprove(address sender, address spender,
                       ElGamalCT E_bob, CPProof pi) view returns (bool);
function verifyIssuerSchnorr(...) / verifyIssuerReenc(...) / verifyDepositorBinding(...);

// The identity accumulator (see The Identity Accumulator)
function setRootAuthority(address next);                             // governance
function enrollSubtree(bytes32 id, uint32 slot, uint8 depth, bool isPublic,
                       address poster, string publication);         // root authority
function evictSubtree(bytes32 id);                                   // root authority
function setAggregator(address next);                               // root authority
function postIdentityRoot(uint256 root, bytes32 leafListHash);       // aggregator
function setMaxRootAge(bytes32 consumer, uint32 maxAge);            // governance
function acceptsRoot(uint256 root, bytes32 consumer) view returns (bool);
function verifyPublicMembership(bytes32 id, uint256 leaf, uint256[] subSiblings,
                                uint256 subIndex, uint256[] aggSiblings, uint256 root)
                                view returns (bool);
function markApproved(address spender);                             // Buck-only
function setBuck(address buck);                                     // governance, one-shot

```

A few non-obvious points:

- `verifyApprove` reads `E_alice` from registry storage so the caller cannot substitute a fake credential. Its transcript binds the verifying registry address, so a proof prepared for a different registry deployment is rejected. Here "registry" means `address(this)` while the proof is verified: for a Diamond facet that is the stable Diamond proxy, not the replaceable facet

implementation. Replacing a facet therefore does not invalidate proofs. Moving the registry to a different proxy requires unsubmitted registration/approve proofs to be regenerated (and registry state to be migrated), but does not alter already minted Notes or their receipts; those artifacts do not contain either proof.

- **bindContract** requires both certification and target control. The compact overload copies a registered caller’s exact stored identity after the target authorizes the full binder/identity/policy tuple. The credential overload verifies PS certification plus holder authorization; target authorization remains mandatory on that path. Contracts with known provenance use a governance-approved, contract-specific adapter. The adapter proves canonical provenance and authority, while the registry copies the named operator’s stored identity rather than accepting adapter-supplied keys. The first authorized binding is immutable.
- **markApproved** (called by Buck inside **approve**) freezes the spender’s **isCarrying** flag the first time anyone successfully approves them, so the carrying flavor a sender consents to cannot be retroactively flipped.
- Revocation does not un-register existing accounts; they were verified once and are never re-verified.

4.3 Public-Identity Contracts

DeFi infrastructure – AMM pools, vaults, oracles, the Notes pool – can’t generate Chaum-Pedersen proofs at **approve** time because the contract has no private key. Such contracts authorize and bind a certified operator identity, or use an approved adapter with audited provenance and authority semantics, with **isPublicIdentity_=true**, **isCarrying=true**. Counterparties of a Public-Identity contract still call the identity-bound **approve** (the always-CP rule applies regardless of spender flavor), but the receipt-fragment check on the contract side is satisfied by the deterministic **_identityHash** rather than a per-counterparty CP receipt. Public-Identity contracts forfeit content privacy: their identity M is openable on subpoena via the operator’s off-chain attestation. Encrypted-Identity contract binding (**isPublicIdentity_=false**) is supported by the surface but not yet used in production.

4.4 Issuer Trust

trustedIssuers is the only governance-controlled trust anchor. Adding an issuer admits a new origin of registrable identities; removing one stops new registrations under that issuer. Existing registrations are unaffected – they were verified at registration and never re-verified. Whether a registered identity is in good standing is a separate, revocable fact: membership in its registry’s subtree of the identity accumulator, which every consumer that needs liveness proves against a posted root of bounded age – see Identity: Liveness Is Membership.

4.5 The Identity Accumulator

Governance appoints a **rootAuthority** (**setRootAuthority**), which enrolls each certifying authority’s subtrees at aggregator slots (**enrollSubtree(id, slot, depth, isPublic, poster, publication)**, **evictSubtree**) and appoints the **aggregator** (**setAggregator**). The aggregator alone posts roots: **postIdentityRoot(root, leafListHash)** writes **identityRoot** and records the root and its posting time in a ring of **ROOT_RING_SIZE = 256**, evicting the oldest. A consumer names itself by a **v2** tag and **acceptsRoot(root, consumer)** accepts a retained root posted within that consumer’s

maximum age (`setMaxRootAge`; Notes seven days, the insurer gate one day, never longer than the full ring covers). Paths are `MEMBERSHIP_PATH_DEPTH = 32` levels: a subtree of 12 under an aggregator of 20. `verifyPublicMembership(id, leaf, subSiblings, subIndex, aggSiblings, root)` checks a public subtree’s path on chain, taking the aggregator index from the subtree’s enrolled slot rather than from the caller, and `publicIdentityLeaf(M)` is the tagged unsalted leaf. Private (salted) membership is proven in zero knowledge inside the Notes gates. `register` and `bindContract` refuse a caller-supplied leaf: admission is the authority’s certification, never a by-product of registration.

5 BUCK: ERC-20 Token

BUCK is an identity-bound ERC-20 with NFT-backed credit headroom, mutual-insurance-pool minting, and continuous demurrage. All transfer-path state for an account fits in one storage slot, so a transfer is one `SSTORE` per side. `mint(N)` opens enough NFT-backed insured-asset coverage to deliver N units of *spendable headroom* to the holder (no BUCK is delivered to raw balance) and simultaneously transfers a *pool principal* from the holder’s signed raw to `insurancePool`, sized so that at the insurer’s assumed annual ROI the principal’s investment yield exactly covers the annual premium on the activated coverage. The holder spends the headroom by transferring out, which drives their signed raw negative; `burn(N)` unwinds coverage most-expensive-first and refunds principal back to the holder, lifting their signed raw toward zero. Every transfer routes through the `IdentityRegistry` and emits `BuckTransferReceipt(from, to, amount, fromHash, toHash)` – ciphertext hashes, no plaintext identity on chain.

5.1 Storage Layout

The transfer-path state for an account is one slot:

```
struct AccountState {
    BuckQty      balance;           // int80 underlying; signed. Range = [-6.04e23, +6.04e23] raw
                                     // = [-6.04e17, +6.04e17] BUCK at 6 decimals. Negative ==
                                     // NFT-backed used credit; positive == held BUCK.
    BuckSeconds  buckSeconds;      // uint120 underlying; |balance| * dt, read by sign:
                                     // fee-seconds while positive, issuance-seconds (the
                                     // lien over time) while negative, 0 at zero (I1)
    uint40       timestamp;        // last crystallisation (safe past year 36800)
    uint16       flags;            // bit 0: demurrage routed to demurragePayer[a].
                                     // bits 1-15 reserved.
}
mapping(address => AccountState) internal _state;
```

`BuckQty` is a user-defined value type wrapping `int80`; constructors `toBuckQty(uint256)` (bounds-checks against the positive cap) and `toBuckQtySigned(int256)` (bounds-checks both endpoints) live in `BuckTypes.sol` so a future width change propagates to `Buck` and `BuckCredit` in lockstep. `BuckCredit` always reads via `asUInt()` (its face, floor, and activated values are non-negative by construction); `Buck` reads via `asInt()` in signed contexts and `asUInt()` only after verifying non-negativity.

A bit in `flags` is the cheapest state this contract has: the word rides in the slot the transfer path already loads and stores, so testing one is free. That is exactly why it is spent only on hot-path

dispatch. Cold-path bookkeeping – the count of accounts a demurrage payer carries – lives in its own mapping instead.

A balance changes sign only at a write, after `_crystallize` has folded the old side’s rectangle through the current block, so the old side’s seconds are settled there (a fee realized, or relief paid) and the new side’s start at zero. Invariant I1 – a zero balance holds no seconds – is kept by every writer that can produce a zero, so the field’s meaning is given by the sign alone.

The slots, grouped by concern:

slot	variable	what it holds
0	<code>_state</code>	the packed per-account word above
1	<code>_totalSupply</code>	sum of the positive balances, the Jubilee fund’s excluded
2	<code>_allowances</code>	ERC-20 allowances
3	<code>_receiptFragments</code>	the identity-bound approve’s receipt per (from, to)
4	<code>mintsBacked</code>	coverage outstanding per credit, in face units (= its activated value)
5	<code>mintsPrincipal</code>	the insurance deposit held against it
6	<code>_jubileeLastUpdate</code>	through when the fund has accrued
7	<code>feesRealized</code>	fees taken out of circulation, cumulative
8	<code>reliefRealized</code>	relief paid out of the fund, cumulative
9	<code>demurragePayer</code>	an account’s accepted fee payer
10	<code>demurragePayerRequest</code>	a pending payer election
11	<code>sponseeCount</code>	how many accounts a payer carries

Tests that seed state by slot number (where the public path would need the identity machinery) go through `test/harness/BuckSlots.sol`, and `test/BuckSlots.t.sol` pins each slot against the live layout. **Nothing derived is stored.** An account’s credit limit, and therefore its `balanceOf`, is recomputed from live `BuckCredit` state on every read; there is no cached copy of it anywhere. `mintsPrincipal` is not an exception – it is a record of what was actually paid, which no amount of present-day state can reconstruct once the appraisal or the rate has moved.

BUCK implements `IERC20` and `IERC20Metadata` directly – no `OpenZeppelin ERC20` inheritance, so the slot layout is explicit and the transfer paths are linear (no `_update` override re-entry). Decimals are 6, matching USDC.

5.2 Signed Balances and `balanceOf`

`signedRawBalanceOf(a)` returns the raw `int80` directly; `rawBalanceOf(a)` clamps negatives to zero for legacy ERC-20-style readers; `signedBalanceOf(a)` subtracts demurrage from positive raw (returns raw directly for negative raw, since used credit accrues no demurrage). `balanceOf(a)` – the ERC-20 view – answers *"what alice can spend right now"*:

```
held(a)          = max(0, signedRaw(a) - feeOwing(a))           // Non-Carrying
used(a)          = max(0, -signedRaw(a))
unusedCredit(a)  = max(0, creditLimit(a) - used(a))
balanceOf(a)     = held(a) + unusedCredit(a)                   // Non-Carrying
balanceOf(a)     = max(0, signedRaw(a))                        // Carrying
```

Carrying accounts (AMM pools, Notes pool, Jubilee) hold no `BuckCredit` NFTs, so `creditLimit` == 0 and the signed-balance machinery degenerates to the prior unsigned model. `feeOwing` and `balanceOfFees` clamp to zero on negative raw – demurrage is a per-second property of **held** BUCK, not used credit.

5.2.1 The Credit Limit Is Read Live

`creditLimit(a) = totalCurrentValue(a) * currentBuckK / 1e18` is computed from scratch on every read. This is the point of `BUCK_CREDIT`, not an oversight: a holder's spendable balance is meant to move when they acquire a credit, when an insurer reappraises one, when an asset depreciates on its schedule, and when `BUCK_K` moves under the whole system. A `balanceOf` that did not move with those would be reporting the wrong number.

An earlier draft cached the product per block, invalidated by an NFT-mutation hook from `BuckCredit`. That could not have been made correct. `BUCK_K` moves inside any `mint` or `burn` that advances the PID – from any account, with nothing to announce it – so two transactions in one block were enough to read a stale limit; and no hook can cover depreciation at all, which is continuous. The cache never populated in practice, and it was removed along with the hook rather than repaired.

The cost lands where it should:

<code>creditLimit(a)</code> for ...	gas
an account with no credits	6,327
an account with 10 credits, <code>NONE</code> schedule	27,603
one credit, 40 years into <code>DECLINING_BALANCE</code>	11,925

An account holding ten insured assets pays for ten insured assets on every outbound transfer. An account with a plain positive balance pays one external call that returns zero, and a Carrying account never reaches the credit path at all. That distribution is deliberate: drawing on insured collateral is a more expensive operation than spending cash, and it should read that way on the gas meter.

5.3 `mint(N)` and `burn(N)`

`mint(uint256 amount)` and `mint(uint256 amount, uint256[] calldata tokenIds)` are the two overloads. The no-arg form runs an internal cheapest-first selector (insertion sort over the caller's NFT list – typically a handful) to minimise premium cost; the second form takes a caller-supplied order an off-chain optimizer pre-computes against effective rate, depreciation, or coverage midpoint.

amount is spendable, at the K the mint runs at. `mint(amount)` raises the caller's `balanceOf` by **amount**: it activates whatever coverage, and pays whatever deposit, that takes at today's `BUCK_K`. A caller who wants 50,000 to spend asks for 50,000, whatever K is; the coverage it activates is the dependent quantity, larger when K is lower. A mint for more than the listed credits can give reverts ("insufficient credit allocation"), with one exception: `mint(type(uint256).max)` delivers everything they can give, activating their whole faces. That is how a holder opens all of a credit at once – a SimLP seeding a pool, or a `BuckBasketEquity` opening its marked credit – without computing `K x face` first and racing K to the block.

`burn(uint256 amount)` mirrors the overload pair: it lowers the caller's spendable by **amount** at the K it runs at, releasing coverage and refunding its deposit. It moves no held `BUCK`; the caller must still be within their limit afterwards. The default selector walks the holder's NFTs **most-expensive-first** – the dearest insurance is released first, returning the largest pool principal per `BUCK` released and freeing expensive coverage capacity for re-use. The asymmetry is not arbitrageable: at one K the per-NFT inversion is symmetric (the same K – e on both sides), so a mint-burn round-trip on the same NFT restores both its `mintsBacked` and `activatedValue`, to rounding.

5.3.1 The Per-Credit Inversion

Two quantities have to be kept apart, and the whole inversion turns on the difference.

`activatedValue` is denominated in **face** units – a slice of the asset *as appraised at issue*. What is actually insured today is that slice scaled by `rho = depreciatedFace / faceValue`. That is proportional coinsurance, and it is the right shape for a depreciating asset: the holder’s share of the asset stays constant while its worth tracks the asset. `creditLimit` reads the scaled figure, so the allocator inverts in **present insured value** and grosses back up into face units at the end.

Each credit carries an annual `premiumRate` in basis points, and **the premium is charged on what is actually insured** – the present value, not the face slice. You pay for the cover you have, not for the cover the asset used to be worth. A unit of present insured value raises the credit limit by `K` and costs a deposit of `e = r * POOL_ROI_INV / BP`, so it yields `K - e` of spendable. Given a target spendable `amount`, rate `r` and today’s `K`:

<code>e</code>	<code>= r * POOL_ROI_INV / BP</code>	the deposit per unit of cover ($K > e$, or the credit i
<code>V</code>	<code>= ceil(amount / (K - e))</code>	present insured value that yields ‘amount’
<code>principal</code>	<code>= floor(K * V) - amount</code>	the deposit, $\sim e * V$
<code>take</code>	<code>= ceil(V * faceValue / depreciatedFace)</code>	face units that carry <code>V</code>

At `POOL_ROI_INV = 10` the pool deposit is ten times the annual premium on `V`, so at the insurer’s assumed 10 % ROI its yield funds that premium in perpetuity; the insurer keeps any return above 10 % as profit. When `rho = 1` the last line is the identity and this reduces exactly to the undepreciated case; at `K = 1` it is `V = amount / (1 - e)`, `principal = V - amount`. A credit whose deposit rate reaches `K` ($e \geq K$: a premium of 750bp at `K 0.75`) can deliver nothing and is skipped.

Charging the premium on present value is not merely fairer, it is the only formulation that does not fall over. Charge on the face slice instead and the inversion becomes `take = amount / (rho * K - e)`, which is unsatisfiable once `rho * K <= e`: a 200bp credit would become unmintable at *any* price below 20 % of face at `K = 1`, because depreciation had eaten the premium margin. With the premium on `V`, `K - e` is independent of `rho`, every credit with a non-zero appraisal still works, and the cost per BUCK delivered stays `e / (K - e)` however old the asset is. That last property is why cheapest-first by `premiumRate` remains the correct selector: depreciation changes how much **capacity** a draw consumes, never how much it **costs**.

The allocator (`_allocateMint`) walks `BuckCredit.batchCreditInfo(tokenIds)` once, then iterates in memory. Per credit the available present value is `capV = (faceValue - mintsBacked) * rho` and the spendable it can yield is `capV * (K - e)`; the credit is either fully consumed or partially consumed via the closed form above to land exactly on `remaining`. A credit whose appraisal has reached zero insures nothing and is skipped. For each credit touched, `_allocateMint` calls `BuckCredit.activateFromBuck(tid, holder, take_i)` to grow `activatedValue` and `mintsBacked` in lockstep – the activation and the principal payment are the same atomic operation.

5.3.2 Releasing: the Deposit Comes Back

The release side is deliberately **not** the mirror of the draw.

Drawing prices coverage: how much deposit does this much cover cost, at today’s appraisal and today’s rate. Releasing does not re-price anything. It hands back the deposit the coverage is carrying, pro rata on the face units let go:

<code>capV</code>	<code>= mintsBacked * rho</code>	present value of the outstanding cover
<code>netCap</code>	<code>= K * capV - mintsPrincipal</code>	spendable it can release: its limit less the depo

```
units = ceil(amount * mintsBacked / netCap)    face units to deactivate
refund = mintsPrincipal * units / mintsBacked  the deposit those units carry
```

The distinction only shows once the appraisal moves, and then it is the whole point. A holder who buys cover at one appraisal and releases it at a lower one gets **their whole deposit back**, not the fraction the shrunken cover would cost today.

That follows from what the deposit is. The pool’s yield on it funds the premium in perpetuity – that is what makes a policy a one-time purchase rather than a recurring expense – so what the holder actually pays for cover is the *opportunity cost* of the deposit, not the deposit. Re-pricing on the way out would charge them twice for one decline: the pool has already been compensated for holding an over-sized deposit against shrinking cover, because it earned its assumed ROI on the full deposit the whole time while owing premium only on what was still insured. Keeping the surplus principal as well would be helping itself twice.

The deposit is stored, in `mintsPrincipal[tokenId]`, rather than recomputed. It cannot be recovered from present-day state: it is the sum over past draws of $V_i * \text{effRate}_i / \text{BP}$, and both the appraisal and the premium rate may have moved between them, so `mintsPrincipal / mintsBacked` is a weighted average with no closed form.

Two properties fall out. $K * \text{capV} \leq \text{mintsPrincipal}$ means the position has no spendable left in it – the holder is underwater and must repay before releasing, which is the same rule the post-burn solvency check enforces, reached from the other direction. And when the holder holds no loose BUCK, $\text{netCap} == \text{creditLimit} - \text{used} == \text{balanceOf}$: **burning exactly their spendable closes the position and squares the deposit**. A mint and an immediate burn of the same amount cancel exactly; partial releases floor, so the dust stays in the pool until the final release hands it back.

Per-NFT bookkeeping: `mintsBacked[tokenId]` tracks the cumulative outstanding allocation against each NFT, capped at `faceValue`. The invariant `mintsBacked[tid] == activatedValue[tid]` holds at every observation point, kept by two rules: activation happens only inside `_allocateMint` (there is no public `BuckCredit.activate()`), and `updateCredit` may not reappraise below activated coverage. They grow together on mint and shrink together on burn. The burn allocator sizes its unwind from $\min(\text{mintsBacked}, \text{activatedValue})$ anyway – with the rules above they cannot differ, and the clamp is there so that any future divergence costs the system a rounding rather than costing a holder their exit. A mint-burn round-trip on the same NFT order at one K is rate-neutral and not arbitrageable – both sides hit the same $K - e$.

The shipped mint sequence:

1. `isVerified(msg.sender)` guard.
2. `buckK.compute()` – advances the PID if `dT` has elapsed; otherwise returns the cached value. Mint and burn together amortize PID work across user activity.
3. Capture `fundingFactor` and – when it is non-zero – the minter’s **pre-activation balanceOf**. The reserve is measured before this mint activates the credit it pledges, so the freshly-activated headroom cannot satisfy its own reserve.
4. `_allocateMint(amount, tokenIds, K)` reads `BuckCredit.batchCreditInfo(tokenIds)` once, walks the slice array, calls `BuckCredit.activateFromBuck(tid, holder, take_i)` per NFT (grows `activatedValue` and `mintsBacked` in lockstep), returns `(totalCoverage, poolPrincipal)`.
5. **Counter-cyclical funding-factor reserve**, enforced now that `poolPrincipal` is known:

```
require(preActivationBalanceOf >= poolPrincipal * fundingFactor / 1e18,
        "BUCK: insufficient mint funding")
```

The reserve scales with the **insurance principal this mint pays into the pool** – NOT the gross BUCK minted, nor the credit limit applied for. Its job is to put a gentle base-level demand under BUCK exactly when many credit-holders come online to mint (each must hold a little BUCK sized to its insurance cost), not to impede issuance – which a gate on the whole credit limit would. A **zero-premium** credit yields `poolPrincipal = 0` (= "zero-cost insurance"), so the requirement is zero and the mint is exempt (the basket / LP bootstrap mints freely). On a non-Carrying account `balanceOf = held + unusedCredit`; `fundingFactor` is 0 under the static controller (gate disabled), ~ 1.0 at parity, and rises above 1.0 when BUCK is undervalued, so it bites hardest on bootstrapping **insured** credit from a thin reserve while the market already signals too much BUCK. Its schedule and its precedents are in The Funding-Factor Reserve.

6. `_accrueJubilee()`, crystallise `insurancePool` and `msg.sender`, debit `poolPrincipal` from the sender's signed raw (`_subBalance` is signed-aware – the raw goes negative if the sender had no positive holdings), credit `poolPrincipal` to `insurancePool`. `_totalSupply` is maintained automatically by `_setBalanceSigned` so that `_totalSupply == sum_a max(0, signedRaw(a))` holds.
7. `Emit Transfer(0, insurancePool, poolPrincipal), Minted(sender, totalCoverage, poolPrincipal, creditValue, buckKValue, newLimit)`.

End-state arithmetic for a fresh mint against a zero-balance, non-credit-using holder:

<code>signedRawBalanceOf(holder)</code>	<code>= -poolPrincipal</code>	(used = principal paid to pool)
<code>creditLimit(holder)</code>	<code>= K * sum V_i</code>	(= K x the activated cover's present value)
<code>balanceOf(holder)</code>	<code>= creditLimit - used</code>	<code>= sum (K * V_i - principal_i)</code>
		<code>= sum amount_i</code>
		<code>= amount</code> (modulo rounding)
<code>totalSupply</code>	<code>+= poolPrincipal</code>	(only the pool's positive contribution grows)

A depreciated credit delivers in full, and so does a low K. The allocator inverts in present insured value at today's K and grosses the face units back up by `faceValue / depreciatedFace`, so the last line above holds whatever the appraisal and the controller have done. Depreciation and a low K cost the holder *capacity* on the credit – the same spendable consumes more face units – never spendable on the mint.

Worked through on a 300,000 structure, LINEAR 200bp/yr on a 60,000 floor, three years in (appraisal 285,600, so $\rho = 0.952$), premium 200bp ($e = 0.2$), at $K = 0.75$:

<code>mint(50,000)</code>	<code>-> V</code>	<code>= 90,909.090910</code>	present insured value, <code>ceil(50,000 / (0.75 - 0.2))</code>
	<code>principal</code>	<code>= 18,181.818182</code>	<code>= floor(0.75 * V) - 50,000</code> (= $0.2 * V$)
	<code>take</code>	<code>= 95,492.742553</code>	face units, <code>ceil(V * 300,000/285,600)</code>
	<code>balanceOf</code>	<code>= 50,000</code>	exactly what was asked for

The premium is charged on the 90,909 actually insured, not on the 95,493 face slice that carries it: at the assumed 10 % ROI the 18,182 deposit yields 1,818/yr, which is 200bp of 90,909. Charged on the face slice it would have been 1,910/yr, and the holder would have been paying for cover the asset no longer has.

`burn(amount[, tokenIds])` mirrors the math – spendable falls by `amount` at the `K` it runs at – but with the most-expensive-first default selector: unwind dearest NFT first, call `BuckCredit.deactivateFromBuckholder, unwind_i` which shrinks `activatedValue` and `mintsBacked` in lockstep, debit `poolRefund` from `insurancePool`, credit `poolRefund` to the holder’s signed raw (climbs toward zero from used credit, or further positive from a positive balance). `quoteMint` and `quoteBurn` are read-only views that return `(totalCoverage, poolPrincipal)` or `(totalUnwind, poolRefund)` for the same allocator math at today’s stored `K`, so an off-chain optimizer can size required limits and refunds before submitting. (The mint itself runs at the `K compute()` returns, which can step if a PID interval has elapsed.)

5.3.3 The Funding-Factor Reserve, and Its Precedents

Step 5 above asks a minter to already hold `poolPrincipal * fundingFactor` before a mint succeeds, where the controller’s factor is counter-cyclical in BUCK’s deviation from the basket:

```
factor = max(0, 1 + 10 * (basket - BUCK) / basket)
```

BUCK price (basket = 1.00)	Factor	Reserve, as a share of the principal this mint pays
1.10	0.0	none – BUCK is scarce, issuance is welcome
1.05	0.5	half
1.00	1.0	the whole principal
0.95	1.5	one and a half times
0.90	2.0	twice

The reserve is a check, not a deposit. It is measured on the pre-activation `balanceOf` – held BUCK plus unused credit from earlier activations – and nothing locks it afterwards. A first-time minter has no unused credit, so it must acquire BUCK first, and that purchase is demand for BUCK sized to the insurance it is about to buy, largest when the market already says there is too much BUCK. A zero-premium credit pays no principal and so needs no reserve, which is how the supply is seeded. Treating the reserve as permanently set aside is a strategy, not a rule, and a poor one: in the equilibrium simulations, borrowers that never released it starved their own later issuance until supply froze, and borrowers that let it go as their positions retired kept the loop closed (`alberta_buck/sim/EQUILIBRIUM.md`).

The mechanism is old; only its trigger is new.

- **Initial margin.** Opening a leveraged position requires posting cash as a fraction of the notional, and exchanges raise that fraction in stressed markets – a counter-cyclical brake on credit expansion. The factor rises the same way when BUCK trades below the basket, but it moves with the market price rather than by an exchange’s decision.
- **Reserve requirements.** Central banks have required commercial banks to hold a fraction of deposits in reserve, raising it to tighten credit creation and lowering it to loosen. The reserve here applies the same lever to the individual minter rather than to an institution.
- **Gesell’s stamp scrip.** *Freigeld* required monthly stamps, bought in the same currency, to keep a note valid, which made the currency its own continuous source of demand. BUCK’s demurrage is Gesell’s direct heir; the reserve is its issuance-side cousin, demand that arises from the act of creating credit.

- **Friendly-society buy-ins.** Mutual-aid societies, and the cooperative banks descended from them, required new members to pay in before drawing benefits. A minter is by construction a contributor to the mutual insurance pool – the principal it pays in – and the reserve is sized to that contribution.
- **Tax-driven demand.** The chartalist account holds that a currency is valued because taxes must be paid in it. The reserve is a structural sink of the same kind: whoever wants BUCK credit must first hold BUCK, in proportion to the system’s monetary stress.
- **Token pre-sales.** Young networks gate participation on pre-purchased tokens. The reserve aligns new entrants the same way without a one-shot sale, and the entry price is never paid away: it stays in the holder’s balance.

5.3.4 Underwater Is a State, Not a Failure

The burn-side solvency check: after refund and deactivation, the holder’s remaining used credit must still fit under the shrunken `creditLimit`, else the burn reverts with BUCK: `post-burn credit used exceeds limit`.

Releasing `A` units of headroom gives back $A/(1-\text{eff})$ of coverage but refunds only the difference, so the limit always falls faster than the debt does. A holder standing exactly at their limit – the normal end state of mint-then-spend – therefore cannot burn at all until they repay. That is the intended order of operations, and the same one any loan imposes: repay the credit, then release the collateral. Acquiring BUCK and receiving it *is* the repayment; it lifts the signed raw toward zero, and the unwind then fits.

A holder can also be pushed under the line by events they did not choose – a reappraisal, or simply the insured asset depreciating on schedule. Doing nothing about it is a supported outcome, and costs them nothing acutely:

- **The insurance stays in force.** Its premium was paid up front, in perpetuity, and no path cancels it. `activatedValue` does not move.
- **The exposure shrinks on its own.** Jubilee relief accrues against the lien – the BUCK the credit actually put into circulation – at $\sim 2\%$ /year, so what closing the position eventually costs declines year on year with no action from the holder. Over ten years a 1,000 BUCK lien’s `redeemCost` falls to ~ 800 BUCK. Undrawn credit earns nothing.

So there is no liquidation, no margin call, and no forced sale. The position simply becomes cheaper to close the longer it is carried – which is the whole point of the Jubilee.

The `balanceOf(sender) > amount = spendable` check on transfer-out is where credit headroom becomes a real BUCK flow: a transfer larger than `held` but at-or-below `held + unusedCredit` drives the sender’s signed raw negative, and the recipient receives freshly-issued BUCK. `_setBalanceSigned` grows `_totalSupply` by exactly the amount of the positive contribution created on the recipient side (offsetting nothing on the sender side, whose positive contribution just shrank from `held` toward zero or further into used credit).

5.4 Identity-Bound `approve()`

`approve(spender, amount, E_bob, pi_CP)` is the identity-bound form, and the only one that lays down a receipt fragment. Both `sender` and `spender` must be `isVerified`. The CP proof is required regardless of the spender’s identity flavor (Public-Identity or Encrypted): the always-CP

rule produces a uniform per-pair audit record openable on subpoena, and engages the spender’s BN254 identity key for forward security under future identity rotation. Full rationale in *Why approve Is Always CP-Bound*.

On success, `_receiptFragments[sender][spender] = keccak256(E_bob)`, the standard ERC-20 allowance is written, and `markApproved(spender)` freezes the spender’s `isCarrying` flag in the registry (so the carrying flavor consented to cannot retroactively flip). Per-pair cost is ~97K gas (29K CP verify + 46K allowance + 22K receipt SSTORE), paid once. Subsequent transfers between the same pair add only the bilateral check and the receipt event (~5K above an ERC-20-baseline transfer).

The plain 2-arg `approve(address, uint256)` is *also* supported, and sets nothing but the allowance. This is deliberate, and it costs no privacy: identity is enforced at **transfer** time, never at approve time. A bare allowance only authorises the spender to *initiate* a `transferFrom`; `_identityCheckedTransfer` then independently re-validates both `from` and `to` and the receipt-fragment rule for that pair – and that gate keys on `(from, to)`, never on `(from, spender)`. A plain-approved spender therefore cannot manufacture a transfer the rules would not already permit: they still cannot move BUCK between two non-public parties without a real CP receipt fragment laid down by the 4-arg form.

What it buys is that BUCK is a first-class citizen of stock router and Permit2 infrastructure, which requires a plain `approve(PERMIT2, max)`, without weakening any counterparty-privacy invariant.

5.5 `transfer()` and `transferFrom()`

Both route through `_identityCheckedTransfer`, which enforces *mutual decryptability*: every private (non-public) party must hold a per-pair CP receipt fragment for its counterparty, established before the transfer by the 4-arg identity-bound `approve`. Only public-identity contracts may use the deterministic `_identityHash` fallback.

- Requires `isVerified(from)` and `isVerified(to)`.
- For the `to` side (`_receiptFragments[from][to]`): if zero, requires `=isPublicIdentity(from)` — reverts with BUCK: sender must identity-approve recipient otherwise. Falls back to `_identityHash(to)` when `from` is public.
- For the `from` side (`_receiptFragments[to][from]`): if zero, requires `=isPublicIdentity(to)` — reverts with BUCK: recipient must identity-approve sender otherwise. Falls back to `_identityHash(from)` when `to` is public.
- Both guards pass only when *both* parties are public-identity contracts. EOA-to-EOA transfers require bilateral CP-approve; EOA-to-Public and Public-to-EOA require the private party to CP-approve the public one.

After identity checks, transfer dispatch is automatic on the sender’s `isCarrying` flag: Carrying senders (AMM pools, the Notes pool, the Jubilee fund) take the `_carryingTransfer` path that crystallises both sides through now and apports the sender’s live `buckSeconds` proportionally to the value transferred; Non-Carrying senders take `_nonCarryingTransfer`, which crystallises the sender, checks the amount against `balanceOf(from) = held + unusedCredit`, and shifts raw balance. Because the sender is crystallised immediately before that check, the spendable figure it reads is always settled through the current block. `BuckTransferReceipt(from, to, amount, fromHash,`

toHash) is emitted; `_identityHash(a) = keccak256(abi.encode(pk, E_addr))` is deterministic per account.

5.5.1 Reentrancy Guard

Every state-mutating entry point – `transfer`, `transferFrom`, the identity-bound `approve`, both `mint` and `burn` overloads, `mintFromBasket`, `burnFromBasket`, and the delegated-demurrage calls – is wrapped in a transient-storage mutex (EIP-1153; the build targets `cancun`). `TSTORE` / `TLOAD` are 100 gas flat with no cold tier, so the guard costs ~ 600 gas per guarded call against $\sim 5,150$ for the classic storage-slot guard, and it occupies **no storage slot** – the layout is undisturbed. Transient state cannot survive the transaction, so a path that sets the lock and returns without clearing it cannot brick anything.

A bit in the spare `AccountState.flags` word would be cheaper still (~ 390 gas, since the slot is written anyway) and was measured and rejected. It guards the wrong thing: an attacker re-enters from whatever address they like, so locking `msg.sender` stops nothing, and locking every account an operation touches means publishing a lock `SSTORE` per account before every call-out. The bit would also share a slot with the balance, and this contract’s idiom is read-struct-into-memory / write-struct-back – every such sequence silently clears a lock set after the read.

The guard has **no exemptions**. It needed one when `BuckCredit` called back into `Buck` through the credit-limit hook – that call arrives from inside the very `activateFromBuck` a guarded mint is executing, so guarding it deadlocked every mint. With the hook gone, the call graph between the contracts runs one way and the question does not arise. The plain 2-arg `approve` is left unguarded on purpose: it touches only the allowance map, calls nothing, and is the hottest entry point for router integration.

5.6 Demurrage and Jubilee

BUCK accrues continuous demurrage to fund a perpetual Jubilee, and the Jubilee pays it back as relief to whoever issued the BUCK: **the relief equals the demurrage accrued**. Each account’s `buckSeconds` field is the cumulative integral of $(|\text{raw balance}| \times dt)$ crystallised at `timestamp`, read by sign: while the balance is positive it counts *fee-seconds*, while it is negative (a lien: credit drawn) it counts *issuance-seconds*. The live fee owed and the live relief accrued are

```
feeOwing(a) = (buckSeconds + balance * (now - timestamp)) * BASE_RATE_PER_SEC / SCALE    balance
reliefOf(a) = min(lien, (buckSeconds + lien * (now - timestamp)) * BASE_RATE_PER_SEC / SCALE)
                                                    lien = -balance > 0
```

with `BASE_RATE_PER_YEAR = 0.02` (2%/year) and `SCALE = 1e27`. Demurrage is a per-second property of *held* BUCK; relief is a per-second property of *issued* BUCK. One field serves both because an account never holds a fee when it crosses zero: **every BUCK carries its fee to the end**, and the fee is *realized* – taken out of circulation – whenever the BUCK carrying it would otherwise shed it:

- **Spending past held BUCK into credit.** The fee locked in the balance is paid first; the lien absorbs it, so the lien is exactly the credit drawn.
- **Aged BUCK arriving at an account below zero.** They repay the lien net of the fee they carry. A receipt that repays the whole lien also pays out the relief the lien earned.

- **A basket burn.** The burned BUCK take their share of the basket’s age with them, and retire that much less of the basket’s issuance.

Relief pays out of the fund when the lien is repaid, when the holder burns coverage, or when the holder calls `settleRelief()` (only the holder: relief accrues linearly on the lien, so paying it early is the holder’s choice). It is capped at the lien, so a lien carried ~50 years closes free. `redeemCost(a) = lien - reliefOf(a)` is the liability-side quote. (Until 2026-09-29 relief accrued in BuckCredit on *activated coverage*, drawn or not – 1/K of what a fully drawn credit issues, and more than the fund took in; doc/JUBILEE-ISSUANCE.org.)

Visible balance:

- **Non-Carrying** (default for EOAs): `balanceOf(a) = held + unusedCredit` where `held = max(0, signedRaw - feeOwing)` and `unusedCredit = max(0, creditLimit - used)`. The locked fee stays inside the account’s own slot; spendable on the held side shrinks over idle periods; unused credit shrinks if BUCK_K falls.
- **Carrying** (AMM pools, Notes, Jubilee, etc.): `balanceOf(a) = max(0, signedRaw)`. The account never loses visible balance to demurrage; instead, on every outflow the value-weighted age is folded into the recipient’s `buckSeconds` so the locked fees ride with the BUCKs. Carrying accounts hold no BuckCredit NFTs and never go negative (the spendable check inside `_carryingTransfer` rejects any transfer that would drive `raw < 0`).

`_crystallize(a)` folds the positive time rectangle into `buckSeconds` and bumps `timestamp`; it never changes balance and tolerates negative signed raw (rectangle = 0). Carrying transfer crystallises both sides through now, apportions `liveBs * value / raw` of the sender’s live integral to the recipient (where `liveBs = buckSeconds + raw * (now - timestamp)`), and settles each side in one SSTORE.

The Jubilee is a normal account at `address(this)`, treated as Carrying. On every mint or burn, and on every transfer that draws or repays credit, `_accrueJubilee` writes

```
_state[address(this)].balance += totalIssued * BASE_RATE_PER_SEC * elapsed / SCALE
totalIssued = totalSupply - reliefRealized + feesRealized      (+ |basketIssued| if negative)
```

`totalIssued` is the BUCK issued – the liens plus the basket’s net issuance – which the relief accrues on, so the fund always holds the relief it owes. It follows from the only three things that move the sum of the non-Jubilee signed balances: the basket’s hooks, relief paid out, and fees realized.

directly into Jubilee’s slot via a raw slot write – bypassing `_setBalanceSigned` – so `totalSupply` is **not** mutated. Demurrage is internal redistribution from positive-balance holders into the system-wide Jubilee, not fresh minting. The exact invariant:

```
sum_a max(0, signedRawBalanceOf(a)) == totalSupply + jubileeActual()
```

Jubilee is the system-wide fee escrow. Every BUCK has exactly one fee owner: non-Carrying accounts lock it silently inside their raw balance (`balanceOf = raw - feeOwing`); Carrying accounts (Notes pool, AMM pools, Jubilee itself) show `balanceOf = raw`, so their fee portion sits in Jubilee instead. When Carrying BUCKs flow to a holder via `_carryingTransfer`, the recipient absorbs `liveBs * value / raw` as extra `buckSeconds`, where `liveBs` is the sender’s full live integral (crystallised history plus the current elapsed rectangle); flowing to an account below zero, that

fee is realized on arrival instead. `JubileeAccrued(delta, newBalance)` surfaces every accrual, `JubileeRedeemed(account, relief)` every payout.

The Jubilee is the sink that funds default coverage, perpetual-ground-rent disbursement, or a periodic balance reset – all governance-driven on-chain disbursement paths land on top of this single accumulating slot.

5.6.1 Delegated Demurrage: a Designated Fee Payer

An account may route its demurrage to a designated payer, so that one Identity's several accounts concentrate their erosion in one place instead of each being eaten from underneath.

The mechanism is a **transfer of buck-seconds**, not a discount. Demurrage here is a lien, never a movement: the fee is locked inside the account's own raw balance, and the aggregate of those liens is what backs the Jubilee's accrual against `totalSupply`. Since `sum_a buckSeconds(a)` tracks `integral(totalSupply dt)`, destroying buck-seconds anywhere would leave the Jubilee accruing against nothing. So at crystallisation the sponsored account's `(balance * dt)` rectangle is moved into the payer's slot rather than its own. The total is conserved exactly, and **no raw BUCK moves at all** – no supply invariant is touched by the feature.

Absorption is capped at the payer's own lien capacity: the point at which `feeOwing(payer)` would exceed `rawBalanceOf(payer)`. Past that the lien would be uncollectible, which is precisely how delegation would have become a way *out* of demurrage rather than a way to relocate it. Whatever the payer cannot carry stays with the sponsored account, so the worst case is "no delegation at all" – never uncollected demurrage.

Consent is two-sided. `requestDemurragePayer(payer)` elects; `acceptDemurragePayer(account)` takes on the liability; either side may `clearDemurragePayer`. The payer's consent is what stops anyone dumping unbounded fee exposure onto any balance. Neither party may be Carrying – a Carrying account's `balanceOf` ignores its fee entirely, so its lien would not bite – and chains are refused, since routing is one hop by construction and a chain only makes "who is actually paying" unanswerable by inspection.

Settlement is lazy, on the sponsored account's next touch, which is the cadence at which BUCK accrues everything else. Between touches the payer's own `feeOwing` does not yet include its sponsees' pending rectangles – finding them would mean enumerating sponsees – so `settleDemurrage(account)` is a permissionless poke that forces the transfer. No spend decision ever uses a stale number: every balance-moving path crystallises the account it is about to debit immediately before reading its balance.

Cost: nothing for accounts that do not opt in, since dispatch is a mask on the `flags` word the transfer path already holds in memory; +863 gas warm and +8,434 cold for those that do. Conservation is asserted against identical unsponsored control pairs; the residual is at most one raw unit (1e-6 BUCK) per account merged, from flooring once over a pooled slot instead of twice, and it always lands on the system's side of the ledger.

5.7 Integrating with BUCK

BUCK is a standards-compliant ERC-20 and works with stock routers, pools and Permit2. Three of its properties are unusual enough that an integration written against ordinary tokens should be checked against them.

Identity is enforced on transfer, and both sides must be registered. A transfer to an unregistered address reverts. Any contract intended to hold BUCK must be bound via `IdentityRegistry.bindContract` – typically with `isPublicIdentity_ = true`, `isCarrying =`

`true` for a pool or vault (see `IdentityRegistry`). This is the one change most integrations need: deploy, then bind, before the first transfer in.

balanceOf moves on its own. For a non-Carrying account it is `held + unusedCredit`, and the credit half is a live function of the holder’s insured assets and of `BUCK_K`. It falls as those assets depreciate, moves when an insurer reappraises one, and moves when the PID moves `BUCK_K` – all without any transfer occurring. Demurrage moves the held half downward continuously for the same reason. Consequences:

- Do not treat a stored `balanceOf` snapshot as still valid in a later block.
- Do not derive a share price or a collateral ratio from an account’s `balanceOf` unless you intend it to track that account’s insured assets. For custody questions use `rawBalanceOf` / `signedRawBalanceOf`, which report only BUCK actually held.
- Do not sample `balanceOf` from inside a callback you do not control: views are not covered by the reentrancy guard, and the ordinary read-only-reentrancy discipline applies.

Contracts bound as **Carrying** are exempt from all of this: `balanceOf` is exactly the raw balance, no credit and no demurrage deduction. Every pool in this system is Carrying, which is why nothing in-tree is affected.

Transfers may exceed balanceOf at the moment you read it, and may exceed held BUCK. A transfer at or below `held + unusedCredit` succeeds by driving the sender’s signed raw negative and issuing fresh BUCK to the recipient. A pre-flight `require(balanceOf(x) >= amt)` in your own contract is therefore redundant, and against a stale read it is wrong in both directions. Let the transfer revert instead.

6 BuckBasket: Direct-Mint Orchestrator + USD-Free Value Anchor

This section describes the pro-rata basket (`BuckBasketProRata` and its heirs `BuckBasketOps`, `BuckBasketFence`, and the legacy `BuckBasket`). It mints and burns BUCK through basket hooks that production Buck does not have; it runs in the simulations, on `BuckWithBasketHooks`, as the baseline the production basket is measured against. The production basket is `BuckBasketEquity`, an ordinary credit holder.

BuckBasket is the contract that closes the loop between BUCK and the real-world commodity tokens it tracks. It registers a set of TOKEN/BUCK Uniswap-V3 pools (one per basket constituent), custodies a single full-range Buck-owned LP position in each, lets anyone direct-mint BUCK by depositing a constituent (or recycle already-minted BUCK into the most underweight constituent), and on redemption pulls BUCK out of overweight pools while routing the surplus into the most underweight pool as treasury LP. It is also the publisher of the USD-free process variable read by `BuckKControllerDirect`.

6.1 Two Kinds of BUCK

In production every BUCK is issued against a lien: drawn on a `BuckCredit` through the standard `Buck.mint` path (BUCK), gated by `BUCK_K`. The equity basket is no exception – its credit is a marked credit on its own equity.

The pro-rata basket is the exception, and only in the simulations:

1. **Hook-minted BUCK** – minted by the basket via `mintFromBasket(to, amount)` when a depositor pledges a TOKEN, tracked in `totalOutstandingBuck`, burned on redemption. The gate is not a credit limit but the depositor's willingness to put TOKEN (or BUCK) in at the pool valuation. `BuckWithBasketHooks` records the hooks' net issuance (`basketIssued`) and the relief it has earned (`basketRelief`, recorded, not paid); a burn's BUCK carry their share of the basket's age out and retire `amount - fee` of that issuance. The fund accrues on it like any lien.
2. **Credit-drawn BUCK** – as in production; these reach the TOKEN/BUCK pools through arbitrage and are the lever the PID pulls on.

The two supply paths converge in the basket's BUCK-side balance. Price-up TOKEN moves leave extra BUCK in the pool (TOKEN bought with BUCK by external traders); price-down moves draw BUCK out. Combined with reinvested treasury BUCK from prior redemptions, the basket's NAV typically exceeds `totalOutstandingBuck` – the spread is what compounds for depositors and underwrites the "profit" side of the redemption split.

6.2 Constituents and Weights

Governance registers each basket constituent via `addBasketToken(token, decimals, initialPriceInBuck, weightBp, feeTier)`. `weightBp` is the declared target weight in basis points; passing 0 defaults to an equal share (10000 / N). The call:

1. Renormalizes every existing constituent's `targetWeightBp` so the total across the N+1 constituents sums to 10000 bp, preserving each existing constituent's **relative** weight ratio.
2. Re-prices every existing constituent's `basketAmount` at the **current** pool spot price so its contribution to `basketValueInBuck` equals its declared weight share of 1.0 BUCK.
3. Finds or creates the BUCK/TOKEN V3 pool at the requested fee tier, initializes it at `initialPriceInBuck` (silently no-ops if already initialized), and bumps the observation cardinality so TWAP reads are available immediately after the first swap.
4. Computes the full-range tick bounds for the pool's fee-tier spacing and stores the constituent's `buckIsToken0` ordering.
5. Calls `controller.reprime()` so the controller absorbs the dilution-discontinuity in `basketValueInBuck` without firing a one-cycle P/I spike.

A future `rebalanceWeights(tokens[], weightBps[])` will let governance re-weight (or, with `weightBp = 0`, remove) one or more constituents in a single call – recomputing all `basketAmount` values from the current spot prices so each constituent's contribution again matches its declared share of 1.0 BUCK.

6.3 Process Variable: `basketValueInBuck()`

```
basketValueInBuck = sum_i ( basketAmount_i * pool_price_in_BUCK_i / 1e18 )
```

This is the direct-embodiment "process variable" – the value of one basket bundle measured in BUCK – and is what `BuckKControllerDirect` reads. TWAP-windowed reads (`twapWindow` seconds) fall back to spot if the pool is too cold to satisfy the request, so the PID degrades gracefully on a freshly-added constituent rather than freezing.

6.4 Direct-Mint: depositToken

`depositToken(token, tokenAmount, maxDeviationBp)` has two paths chosen on `token`:

- **Basket TOKEN deposit**: the depositor pledges a registered constituent. `BuckBasket` reads the pool's spot price, mints `tokenAmount * spotPrice` worth of BUCK to itself via `Buck.mintFromBasket`, and LPs the (TOKEN, BUCK) pair as full-range liquidity in that constituent's pool. An ERC-721 receipt (`BuckBasketReceipt`) records the (BUCK principal, TOKEN principal, original token, deposit time).
- **Already-minted BUCK deposit** (`token = address(buck)=`): the depositor's BUCK is pulled in, swapped on the **most underweight** pool for that pool's TOKEN, partner BUCK is freshly minted against the received TOKEN, and the full (TOKEN, BUCK) pair is LP'd into the underweight pool. This is the cross-pool rebalance arrow on the deposit side.

`maxDeviationBp` optionally guards against TWAP/spot divergence (set to 0 during bootstrap or cold-pool deposits; non-zero rejects the deposit if $|\text{spot} - \text{TWAP}| > \text{maxDeviationBp} / 10000$).

6.5 Redemption: "Sell High, Recycle to Buy Low"

`redeem(receiptId, redeemBp, maxDeviationBp)` settles a value claim `valueClaim = redeemBuck * NAV / totalOutstandingBuck` where NAV is the basket's total LP value across all pools. The claim is split across constituents by `_allocateRedemption`, then settled in seven phases:

1. **Validate + size** the receipt; compute `redeemBuck` as either the full `buckPrincipal` or a basis-point fraction.
2. **Slippage guard** on the depositor's original pool (the known gap: this guard should also fire on every pool actually withdrawn from – tracked as BUG #5).
3. **Allocate** `redeemValue` across pools. Each pool's "ideal" allocation is $\text{alloc}_i = v_i - t_i * (\text{NAV} - \text{redeemValue})$ where t_i is the pool's natural value share at target. Heavily overweight pools contribute *more* than their proportional share; heavily underweight pools clamp to zero. Two passes handle the equilibrium case (proportional) and the rare "must dip into underweight" case. A small-redemption fast path takes the entire claim from the most overweight pool when `redeemValue` is below `SMALL_REDEEM_BP` (1 %) of that pool's value, saving the multi-pool gas in the dominant case.
4. **Burn LP** from each allocated pool. TOKEN and BUCK are held inside `BuckBasket`; no transfers yet, so any aggregate BUCK shortfall (e.g. from a single pool dominating the allocation but carrying mostly TOKEN) can still reach back into the basket's TOKEN holdings.
5. **Cover shortfall** by greedy TOKEN→BUCK swaps on the pools with the most TOKEN balance left, using exact-input swaps padded by `SHORTFALL_BUFFER_BP` (2 %). Any residual gap below `MAX_ORPHAN_DUST_WEI` (1 μ BUCK) is tolerated; larger gaps revert with "pool depth too thin".
6. **Pay depositor** the surviving TOKEN side per pool, emit one `RedeemedFromPool` event per touched pool (TOKEN-to-user, TOKEN-swapped, L burned).

7. **Reinvest profit** above the `MIN_REINVEST_BUCK` floor (0.001 BUCK): the BUCK side minus the principal burn is routed through `_buckToLp` into the most underweight pool’s TOKEN and LP’d as a treasury position. Sub-floor profit stays in the basket’s BUCK balance and is swept by the next redemption (still treasury-owned, just not yet LP’d). Then `Buck.burnFromBasket` burns the principal, `totalOutstandingBuck` decrements, and the deposit is finalized (cleared on full redemption; both BUCK and TOKEN principals scale down proportionally on partial).

The depositor/treasury split is **approximately** 50/50: full-range V3 LP is roughly 50/50 by value at any price, so the TOKEN side (depositor) and BUCK side (treasury) are roughly equal. The principal burn comes out of the BUCK side, slightly favouring the depositor.

6.6 Invariant

After all deposits exit, `totalOutstandingBuck` = 0 and remaining LP value is entirely treasury-owned – compounded TOKEN appreciation, accumulated AMM fees, and external-arb BUCK that didn’t get claimed. Modulo bounded `MAX_ORPHAN_DUST_WEI` per-redemption dust, this is the liveness backstop the audit trail rests on.

6.7 Known Gaps

The contract carries BUG #N: tags for the redesign work still in flight:

- #5 – Slippage guard runs on the depositor’s original pool, not on the pools actually withdrawn from.
- #6 – `_buckToLp` / `_swapTokenForBuckExactIn` swaps use the permissive `MIN/MAX_SQRT_RATIO ± 1` price limit; should be TWAP-bounded to resist sandwich.
- #7 – `_mostUnderweightPool` / `_mostOverweightPool` read spot prices; should use TWAP.
- #8 – `_reinvestBuck` LPs into a single most-underweight pool; should mirror the redemption’s proportional allocation.
- #9 – TOKEN deposits LP into the deposited token’s own pool with no underweight-routing – asymmetric with the BUCK-deposit path.
- #10 – `Buck.mintFromBasket` bypasses `BuckK’s fundingFactor` gate; document or fold basket-minted BUCK into the PID’s accounting (see Dynamic Issuance below).
- #11 – `_reinvestBuck` emits no event; downstream observers can’t distinguish treasury reinvestment from external arb trades.
- #13 – Late-basket-life redemptions can revert `pool depth too thin` when the basket NAV is too low to cover the remaining outstanding (genuine liquidity exhaustion, not dust); a graceful tail-redemption mode is future work.

6.8 Dynamic Issuance: Why BuckBasket Needs BUCK_K

`BuckBasket` mints BUCK at **exactly** the current pool spot price. That is value-neutral at the moment of deposit, but it is also the cause of a structural deflationary pressure on BUCK:

- Direct-mint adds new BUCK only when a depositor **wants** to LP a constituent. External BUCK demand (somebody wants to buy a TOKEN and the cheapest path is BUCK -> TOKEN swap on the BuckBasket pool) does **not** mint new BUCK – it just rearranges existing BUCK between the pool and the swap-caller.
- The only path that produces "free" BUCK against assets **outside** the basket is `Buck.mint(amount, tokenIds)` via `BuckCredit` NFTs, gated by `maxLimit = totalCreditValue * currentBuckK / 1e18`. When external arbitrageurs buy TOKEN with BUCK from the basket pools, BUCK gets scarce (price up vs. TOKEN each TOKEN/BUCK pool quotes a higher TOKEN price in BUCK terms `basketValueInBuck` rises above 1.0).

Sign convention (inherited from BUCK_K):

- `basketValue > 1.0` (BUCK undervalued / inflation: TOKEN expensive in BUCK): `error < 0` `buckK decreases` `BuckCredit` backed mints shrink BUCK supply contracts basket value falls back toward 1.0.
- `basketValue < 1.0` (BUCK overvalued / deflation: TOKEN cheap in BUCK): `error > 0` `buckK increases` `BuckCredit` backed mints expand BUCK supply grows basket value rises back toward 1.0.

So the closed-loop equilibrium picture is: `BuckBasket` establishes the TOKEN-side anchor and publishes the deviation signal; `BuckKControllerDirect` integrates that deviation into a credit-limit multiplier; `Buck` meters new supply against the multiplier; external arb agents close the loop by moving BUCK between the basket pools and the wider market. Direct-mint deposits are the **value injector** (asset coming in); `BuckCredit` backed mints under PID control are the **supply regulator** (BUCK going out).

7 BuckBasketEquity: the Basket as a Credit Holder

The production basket holds shares of equity, one pooled lien, and a wallet its work wheel places. Three contracts share one storage layout: the shell `BuckBasketEquity` (deposit, redeem, the treasury, views), the components facet `BuckBasketEquityWheel` (the wheel's steps), and the venue facet `BuckBasketUniswapV3` (the pools). The shell delegate calls the facets: `wheelDue` / `wheelStep` to the components, every other unknown selector to the venue. A monetary desk, when there is one, is a separate contract beside it (The Desk: Its Own Credit Holder).

The rule the layout follows: **independent machines get independent accounts and independent wheels**. Buck keys everything by address – one signed balance, one lien, one limit, one relief quote per address – so bookkeeping "sub-accounts" inside one contract are invisible to it, and it cannot hold one party within its own collateral. The basket's depositors and the desk are two parties with two books of collateral, so they are two contracts, two BUCK accounts, two marked credits, and two work wheels (one engine, `BasketWheel`, instantiated per machine with only that machine's kinds, and its own reserve and callers). A caller who wants both turned in one transaction batches the calls; nothing is shared to do it.

7.1 The Account and the Credit

The basket is an ordinary BUCK account – bound public and non-Carrying – and an ordinary credit holder. `openCredit` (governance, once) has it issue itself one marked credit: the basket is its insurer, the premium is zero, and the face caps all it may ever draw at $K \times \text{face}$. Before anything that spends – a deposit, an exit, every wheel step – the basket marks the credit at its equity at the exit marks, and nothing else, so Buck enforces `creditLimit(basket) = K \times \text{equity}` – the depositors' equity. The credit is activated, with one `Buck.mint`, at the first mark above zero.

It never mints or burns. Spending past its held BUCK (placing a position, buying TOKEN, paying an exit) draws on the credit; BUCK it receives (a position's BUCK side, a sale's proceeds, a deposit, fees) repay the lien. Its debt is its lien, which earns Jubilee relief like any other; the Daily step collects it.

7.2 Valuation

A position's fair value is twice its BUCK side at a price (full range: $2 L \sqrt{\text{P}}$). Three marks: TWAP (the price), HIGH (each pool at the higher of spot and TWAP – the incumbents' side of an entry) and LOW (the lower – their side of an exit).

```
gross(mark) = sum over pools (position value at mark + idle TOKEN at mark)
equity(mark) = gross(mark) + signedBalance(basket) + reliefAccrued(basket)
```

where `signedBalance` is negative by the lien (or positive by BUCK held, net of its fee). `liquidity` is what Buck lets the account spend – held BUCK plus headroom under the limit. Its target is the larger of 1% of the gross and enough that half the target covers two sigma of the daily net flow times $1 + K$, capped at 20% of the gross.

7.3 Deposit

A deposit of TOKEN or BUCK is equity: valued at its pool's LOW mark (BUCK at par), against the basket at its HIGH marks, less a charge – the pool fee on the swap the wheel will make to place it: $(1 - K)/2$ of the pool fee for TOKEN, $(1 + K)/2$ of the dearest for BUCK. It buys shares, lands

in the wallet (BUCK straight into the account), and raises the mark: $K \times \text{value}$ of new credit. Into an under-water basket (lien above its limit) the new credit first restores the limit – an under-water account stops issuing.

7.4 Redemption

The treasury takes 25% of the receipt's gain over its cost basis, as shares. The rest leaves at the exit marks, less the same charge as a BUCK deposit. The shell first marks the credit at the equity that stays ($\text{equity}(\text{LOW}) - \text{pay}$), then pays in BUCK if Buck lets the account spend it. Otherwise the exit is paid **in kind**: its fraction f of every position and of the wallet's TOKEN, by transfer, with no swap. The BUCK its positions return repay its share of the lien; what they returned beyond that share is paid in BUCK. Two corners: short of its lien share (leverage above one) it leaves TOKEN behind worth the shortfall at the LOW marks; owed more BUCK than an under-water account can spend, the rest stays in the receipt as shares at the post-exit price. No charge in kind.

7.5 The Wheel's Components

Each step is bounded, marks the credit first, and trades at most 1% of a pool's depth (swapCapBp), the arbitrage re-pinning between steps:

- **Daily**: fold the day's net flow into the flow EMA; collect the relief on the lien; the director's sample.
- **Sync(i)**: collect pool i's fees (TOKEN to the wallet, BUCK into the account).
- **Deploy(i)**: pair pool i's waiting TOKEN with its share of the credit beyond the target – pro rata to the TOKEN waiting in every pool – selling part of the TOKEN if the pair is short.
- **Fund**: buy the director's pick (or, past the parking room, the largest gap) with half of what it places; Deploy pairs it.
- **Trim**: when liquidity is below its floor, unwind a slice of a position (the director's pick, else the most overweight) and sell its TOKEN until liquidity is back at its target – including, after a K cut, the amount the basket is over its new limit; or trim what the director names.

7.6 The Director

`EquityTurnDirector` (optional) supplies leaning targets and two gates, per leg, from a ladder of EMAs of each pool's TWAP tick against the basket's mean (The BasketWheel, "The director"):

- **The lean**: the declared weights scaled by $\exp(-\text{tilt} \times \text{excursion})$ and renormalized, the excursion being the leg's deviation from the basket's mean against its anchor EMA (1280 days) – a leg rich against its anchor is leaned down. `tilt` defaults to 1.
- **The turn**: six shorter EMAs (5 to 160 days) each vote when moving back toward the anchor; `quorum` votes (default 4) is a turn. A leg not turned is running.
- **mayTrim(i)**: true unless the leg is still running up – a run is let run – and true regardless once its weight is over its declared target by the leash (`leashBp`, 30% of the target). **mayFund(i)**: the mirror, for a leg running down. The leash is measured against the declared targets, not the leaned ones.

With no director, the basket keeps its declared targets and trims or funds only beyond a plain weight band.

7.7 The Desk: Its Own Credit Holder

The monetary desk (**MonetaryDesk**: the four quadrants, the bounds, the stabilizer seam; The BasketWheel, "The desk") beside an equity basket is **EquityDesk**: its own contract, so its own account at Buck – bound public and non-Carrying – and its own self-issued marked credit, marked before every operation at its own net value: its **TOKEN** at the pools' low marks, plus its account and the relief accrued on it. Its founding grant (**capitalizeMonetary**) is its own **TOKEN**, in its own contract, collateral for it alone. It issues by spending its own credit, so Buck holds its lien within $K \times$ its own value; BUCK it buys repay that lien (their fee on arrival is its own); the relief on its lien is its own, collected before each operation, and melts its outstanding issuance.

From the basket it reads two things: its equity, which sizes the desk's bounds (as the ops shell's NAV sized them), and its constituents, mirrored so the desk trades in the same pools through the same venue facet (the desk's fallback delegatecalls to it). Nothing the desk holds or owes enters the basket's books, mark, or limit. Its invoker is its own: **monetaryOperation** once per director epoch, from a **BasketWheel** carrying only the ops kind (pointed at the desk) or a keeper.

(It was first built as a sub-account of the basket's one account, marked at the basket's equity *plus* the desk's book. The desk's founding grant then raised the depositors' limit, and the savings world levered its depositors to a lien of 6.3M on 5.8M of equity at $K = 0.75$. The separation above is structural for that reason: two accounts, so Buck enforces each limit.)

7.8 Invariants

- E1: **equity** = **gross** + **signedBalance** + **reliefAccrued** (clamped at zero), at any mark.
- E2: every issuance is within $K \times$ **equity(LOW)** of the depositors alone: the credit is marked at that equity, and nothing else, before any spend, and Buck enforces the limit.
- E3: no margin calls: a K cut changes no lien; only the wheel's paced Trim repays.
- E4: an exit takes no more than its fraction: in BUCK, at most its value at the LOW marks less the charge; in kind, exactly **f** of the positions and **TOKEN** with its lien share repaid; the remaining holders' share price is whole, to rounding.
- E5: the wallet's **TOKEN** is backed: **token.balanceOf(basket)** > **idleToken** =.
- E6: the mark after an exit is the equity that stays, read after its positions are burned.

And for the desk beside it:

- D1: the desk spends only after a mark at its own net value, so its lien stays within $K \times$ that value (Buck enforces it on the desk's own account).
- D2: nothing the desk holds or owes enters the basket's books, mark or limit.
- D3: the relief on the desk's lien is the desk's, and melts its outstanding issuance.

8 BUCK_K: Value Stabilization Controller

BUCK_K is the loan-to-value cap on insured credit: a holder's credit limit is its activated credit value times BUCK_K. Lowering it contracts every holder's borrowing capacity at once; raising it expands it. The controller's job is to move that one number so that BUCK holds its value against the commodity basket, and it is deliberately slow: BUCK_K is a structural lever that glides over months, while private demand – arbitrageurs, the basket, holders managing their own positions – does the fast stabilization.

Buck sees every controller through the three-function IBuckK interface: `currentBuckK()`, `compute()` (run a cycle if one is due, and return BUCK_K), and `fundingFactor()` (the counter-cyclical mint reserve of The Funding-Factor Reserve). `Buck.mint` and `Buck.burn` call `compute()`, so user activity drives the controller and pays for its work; the function is also permissionless, so a keeper can advance it through quiet stretches.

8.1 The Loop

The error is the BUCK-side reference minus the basket-side reference, and every shipped gain is positive:

- **BUCK below the basket** (inflation): the error is negative and BUCK_K falls. Credit contracts, holders running near their limit burn, supply shrinks, and BUCK recovers toward the basket.
- **BUCK above the basket** (deflation): the error is positive and BUCK_K rises. Credit expands, holders can mint and sell into the premium, supply grows, and BUCK gives back value.

Sign-convention canary tests in each PID controller's suite fail within one cycle if a gain is ever negated.

The output is a neutral value plus proportional, integral and derivative trims, clamped to governance-set rails. At a rail the integrator may only move back toward the interior, so a railed controller does not wind up. The rails matter economically as well as numerically: maximum system leverage is $1 / (1 - \text{BUCK_K})$, so 1.0 is the infinite-leverage boundary, and an upper rail below it keeps a railed controller solvent and makes "pinned" distinguishable from "settled". The closed-loop simulations run an integral-dominant loop (an integral time of about ninety days) around a neutral 0.75 with rails (0, 0.95).

Cycles are paced. `dT` is the minimum interval between cycles – within it `compute()` is a single cached read, so the mints of one window share one cycle's gas – and `dTMax` treats a long silence as a single bounded step rather than integrating a day of stale error at once. Price references read from pools are time-weighted averages, which is the controller's resistance to single-block manipulation: a flash-loan spike that moves spot to half its value leaves the TWAP unmoved.

Governance tunes the loop without stepping its output. `retune` changes the gains and re-derives the integrator so the next cycle reproduces the current BUCK_K (bumpless transfer); `setRails` moves the rails and clamps the live value into them; `setDT` and `setDTMax` set the pacing.

8.2 Three Controllers

The PID machinery – state, pacing, rails, anti-windup, the funding factor and the governance surface – lives in the abstract `BuckKControllerBase`. A concrete controller supplies only its two references.

Controller	BUCK side	Basket side
<code>BuckKControllerStatic</code>	–	–
<code>BuckKControllerDirect</code>	1 BUCK, by definition	<code>BuckBasket.basketValueInBuck()</code> over <code>TOKE</code>
<code>BuckKController</code>	BUCK/USDx Uniswap V3 TWAP	Chainlink feeds and V3 commodity-pool TWAP

- **Static.** Holds one value that governance sets, and reports a funding factor of zero (the reserve gate disabled). It lets the identity, demurrage and Notes layers run where no price reference exists yet.
- **Direct.** Pairs with `BuckBasket`: the setpoint is one BUCK, and the process is what the basket’s constituents are worth in BUCK in their own `TOKEN/BUCK` pools, with no stablecoin in between. The loop runs in parts per million with gains scaled so the output lands directly in `BUCK_K` units, around a governance-set neutral `buckK0` (itself adjustable bumplessly with `setBuckK0`). Because the setpoint never moves, the derivative acts on the error itself. The basket is wired once (`setBasket`), and `BuckBasket` calls `reprime()` after adding a constituent, so the one-block jump that dilution causes is not read as an error.
- **External-Basket.** The original controller: BUCK’s price and the basket’s cost, both in USD, from separate pools and feeds (the target basket is `XAUT/USDT`, `PAXG/USDC`, `cbBTC/USDC` and `WBTC/USDT`, a quarter each – two golds and two bitcoins against two stablecoins, so no single token or stablecoin freeze breaks it). Its output is 1.0 plus the trims. It is primed at deployment so the first cycle at parity reproduces the deployed value, and its derivative subtracts the basket’s own movement, so a gold spike is not mistaken for BUCK moving. Pool reads go through `UniswapV3OracleLib`, a 0.8-compatible port of the V3 oracle helpers.

9 Notes: Privacy-Preserving Denomination

BUCK transfers carry a public (`from`, `to`, `amount`) quartet. Notes lift values, flavours, and identity bindings off the public ledger by escrowing BUCK in a commitment pool: holders post Poseidon-hash commitments via a `Groth16` mint, then redeem any commitment later via a second `Groth16` spend with a fresh nullifier. The contract’s public state is just a list of opaque 32-byte commitments, a 30-slot recent-roots ring buffer, a nullifier set, and a running face-value total. Full design in `Notes` and `Proofs`; this section documents the Ethereum surface only.

9.1 Mint Circuit (Batch, In-Circuit Merkle Insertion)

`circuits/mint_batch.circom` is a parameterised batch-mint circuit. Each pinned size `N` (specialised at $N \in \{1, 2, 4, 8, 16, 32\}$, with the smaller sizes useful for unit tests and small wallet flows) produces its own `MintBatchN*Groth16Verifier.sol`. The shipped `Notes` contract holds a single `mintVerifier` adapter; per-`N` dispatch lives inside the adapter (a single `IMintVerifier` chooses the right pinned verifier by `cms.length`). The public signals are:

Signal	Description
<code>oldRoot</code>	Live note Merkle root the prover read from chain
<code>newRoot</code>	Root after folding <code>cm[0..N-1]</code> starting at <code>nextLeafIndex</code>
<code>nextLeafIndex</code>	Live tree size the prover read from chain
<code>totalFace</code>	Sum of note values, range-bounded to 128 bits
<code>cm[N]</code>	The commitments themselves, so the calldata is what the proof attests
<code>issuerMode[N]</code>	Per-leaf output: PUBLIC or PRIVATE, from the committed flavour

The private-issuer sibling `mint_batch_a2.circom` (per-N `MintBatchA2N*Groth16Verifier`, behind `MintVerifierA2Adapter` and `setA2MintVerifier`) replaces `issuerMode` with each leaf’s `eIss` and the binding’s `T` as outputs, recomputing `idHash = Poseidon-11(T_ID, eNote, eIss, T)` in circuit: $7N + 4$ public signals, so each A2 leaf’s re-encryption binding is pinned to the leaf it names. `T_CM`, `T_ID` and `T_NF` are the note hashes’ v2 domain tags (`circuits/note_tags.circom`).

Per leaf, the circuit enforces a tagged Poseidon-6 commitment opening `cm_i == Poseidon(T_CM, flavor_i, v_i, rho_i, idHash_i, predicate_i)`, `Num2Bits(128)` range bounds on `v[i]` and `totalFace` (closing field-overflow attacks), one sum check `totalFace == sum v[i]`, and an $N \times \text{TREE_DEPTH}$ Poseidon-2 dual Merkle walk: `priorWalk` consumes `ZERO_VALUE` at the insertion slot to attest `oldRoot`, `postWalk` consumes `cm[i]` to advance `rollingRoot` to `newRoot`. Both walks share prover-supplied siblings. About 21.9K R1CS per leaf, half of it the dual Merkle walk; 22,052 at $N=1$.

A batch of one mints for 513K gas end to end (`NotesE2E`, the public-issuer path with its Schnorr); each further leaf adds its commitment and issuer mode to the Groth16 verification and its calldata, tens of thousands of gas rather than a Poseidon insertion’s ~800K. Promotion to $N \geq 64$ needs one of the three escape hatches in `alberta-buck-notes-rollup-mint.org` (calldata-IC hash-pinned, sharded `SSTORE2`, plain `SLOAD`), with calldata-IC the natural fit for the Holochain wallet.

The trusted setup in `scripts/snark/setup.sh` uses a fixed dev-only entropy string. Production would run a multi-party ceremony per pinned N ; the script is the scaffolding.

9.2 The Spend Proof and the Deposit Gates

`circuits/spend.circom` is the one note proof every spend reuses. Public inputs [`noteRoot`, `nullifier`, `face`, `recipient`, `chainId`, `flavor`, `issuanceCommitment`]; private witness (`flavor`, `v`, `rho`, `idHash`, `predicate`, `pathElements[20]`, `pathIndices[20]`). It proves `cm == Poseidon-6(T_CM, flavor, v, rho, idHash, predicate)` under `noteRoot`, the flavour-agnostic nullifier `Poseidon-3(T_NF, rho, idHash)`, `face == v` with both bounded to 128 bits, and that the public `flavor` is the committed one; for B1 it exposes the commitment as `issuanceCommitment` so `Notes` can authenticate the public issuer, and A1/A2 pass zero. About 12,000 R1CS.

The identity half of a spend is a separate gate, and its shape follows how many secrets the spender’s two claims rest on:

- **Addressed (A1, A2): one folded proof.** `deposit_fold_a1.circom` (3,309,276 constraints, 43 public inputs) and `deposit_fold_a2.circom` (3,833,338, 42) prove over one witness that the receiving secret k opens the note’s ciphertext and the spend’s re-encryption `eEnc`; that the depositor’s registered credential decrypts to the identity m ; that a certified leaf `Poseidon(tag_R, m, k, salt)` has a 32-level path to `identityRoot`; and that the nullifier is the one the note’s `idHash` produces. A2 adds the issuer’s own membership and the key tie $T == r'k * G + \text{gamma} * H_{\text{PEDERSEN}}$ against the T its `idHash` commits. `DepositFoldVerifierAdapter` derives every public input on chain, reading the depositor’s key and credential from the registry at `msg.sender`.

- **Bearer (B1): a sigma and a membership proof.** `IdentityRegistry.verifyDepositorBinding` is an EIP-196 Okamoto sigma binding the payout account, `eDepForIss` (the depositor’s identity encrypted for the public issuer) and $P_{\text{dep}} = M + b \cdot H_{\text{PEDERSEN}}$ to one identity scalar; `identity_membership_b1.circom` (494,701 constraints, 9 public inputs) proves the same P_{dep} opens to a certified salted leaf on a 32-level path. B1’s two claims rest on one secret, so the sigma’s shared challenge is a real tie; the blind is on a generator hashed to the curve, so it cannot be shifted onto another identity.

Every setup is dev entropy. Production runs a ceremony per circuit. The arguments are Proofs Part IV, Theorems 7-12.

9.3 Verifier Interfaces

`Notes` is decoupled from the SNARK toolchain via narrow interfaces:

```
interface IMintVerifier {          // public-issuer batches, per-N dispatch
    function verifyMint(bytes proof, uint256[] issuerMode, uint256 oldRoot,
        uint256 newRoot, uint256 nextLeafIndex, uint256 totalFace,
        uint256[] commitments) external view returns (bool);
}
interface IMintVerifierA2 {        // private-issuer batches: eIss and T per leaf
    function verifyMint(bytes proof, uint256[4][] eIss, uint256[2][] T,
        uint256 oldRoot, uint256 newRoot, uint256 nextLeafIndex,
        uint256 totalFace, uint256[] commitments) external view returns (bool);
}
interface ISpendVerifier {
    function verifySpend(bytes proof, uint256 noteRoot, uint256 nullifier, uint256 face,
        address recipient, uint256 chainId, uint256 flavor,
        uint256 issuanceCommitment) external view returns (bool);
}
interface IDepositFoldVerifier {  // the addressed gates
    function verifyFoldA1(bytes proof, uint256 nullifier, uint256 face, uint256 identityRoot,
        IdentityRegistry.ElGamalCT eEnc, address depositor) external returns (
    function verifyFoldA2(bytes proof, uint256 nullifier, uint256 identityRoot,
        IdentityRegistry.ElGamalCT eEnc, address depositor) external returns (
}
interface IIdentityMembershipVerifier { // B1’s membership over P_dep
    function verifyMembership(bytes proof, uint256 identityRoot, uint256 px, uint256 py)
        external returns (bool);
}
```

Stub implementations ship for unit tests that exercise state transitions unrelated to the SNARK. The production adapters decode the proof as `(uint256[2], uint256[2][2], uint256[2])`, build the public-signals tuple, and forward to the snarkjs-generated Groth16 verifier; the mint adapters dispatch internally by `cms.length` to the pinned per-N verifier.

9.4 Notes Contract

`src/Notes.sol` is constructed with `(buck, mintVerifier, spendVerifier, governance)`; `governance` wires the rest: `setA2MintVerifier`, `setIdentityRegistry`, `setDepositFoldVerifier`

(which refuses `address(0)`: the fold IS the addressed gate, so no configuration leaves a weaker one answering) and `setIdentityMembershipVerifier`. Its state is intentionally lean – per-leaf insertion math lives in the SNARK:

- **Audit / double-spend state:** `mapping(uint256 => bool) nullifiers, uint256 noteFaceSum`.
- **Tree state:** `uint32 nextLeafIndex` and `uint256[ROOT_HISTORY_SIZE=30] roots` with `uint8 currentRootIndex`; the constructor seeds `roots[0]` with `EMPTY_ROOT`, the root of a tree whose empty leaf is the field tag of `AlbertaBuck/Notes/Zero/v2`.
- **Records:** each public-issuer commitment’s issuer, so a B1 spend can authenticate it.

`mint` has two overloads. The public-issuer one takes the batch Schnorr (`verifyIssuerSchnorr` against the caller’s registered key); the private-issuer one takes one `A2Binding` per leaf and checks each against `verifyIssuerReenc` and against the `eIss` and `T` the A2 mint proof exposes for that leaf. Both check the stale-state guards (`oldRoot == roots[currentRootIndex]`, `nextLeafIndex == self.nextLeafIndex`), field-bound each commitment, verify the batch proof, pull `totalFace` BUCK, advance the ring and emit `Minted(issuer, totalFace, startIndex, count, newRoot)`; the commitments live in `calldata` for off-chain replayers.

There is no unbound spend. `spendCoupledA1` and `spendCoupledA2` take (`proof, root, identityRoot, nullifier, face, recipient, eEnc, foldProof`), `spendCoupledB1` takes (`proof, root, identityRoot, nullifier, face, recipient, issuanceCommitment, issuer, eDepForIss, b1Proof, membershipProof`). Each requires `root` in the recent-roots window, a fresh nullifier, the spend proof for its flavour, and `identityRegistry.acceptsRoot(identityRoot, NOTES_MEMBERSHIP_CONSUMER)`, then its deposit gate; it burns the nullifier, decrements `noteFaceSum`, and pays with `buck.transfer(recipient, face)`. Because the pool is bound under the Carrying flag, the transfer takes the carrying path: the recipient absorbs the pool’s average demurrage age via `buckSeconds`, preserving `face == v` at the ERC-20 boundary. The events `SpentCoupledA1`, `SpentCoupledA2` and `SpentCoupledB1` publish no identity point; B1’s carries `eDepForIss`, which only the issuer decrypts.

10 End-to-End Walkthrough

Alice owns a \$500K Calgary home (land \$200K, structure \$300K). Two insurers (LandCo, HomeCo) have issued BUCK_CREDIT NFTs; HomeCo’s structure credit depreciates LINEAR at 200bp/yr against a 20 % floor. Three years on, and before Alice has minted anything:

Token	Insurer	Face Value	Depreciation	Depreciated Face	Premium Rate
#42	LandCo	200,000	NONE	200,000	50bp
#43	HomeCo	300,000	LINEAR, 200bp, 20% fl	285,600	200bp

Note what is *not* in that table. `activatedValue` is zero on both credits, because activation happens only inside `Buck.mint`; so `totalCurrentValue(alice) = 0` and Alice’s `creditLimit` is zero too. The credit limit is not "what she could borrow" – it is "what she has already drawn coverage for". Her **capacity** to mint is bounded per credit by `faceValue - mintsBacked`.

10.1 Step 0: Identity Registration (One-Time)

Alice obtains a PS-signed credential off-chain from a trusted issuer (the Alberta government, ATB Financial, etc.), blinds it into a presentation (`A`, `B`), generates a fresh ElGamal key pair, encrypts

her identity point $M = m \cdot G$ under her public key, and produces the Schnorr-family NIZK binding the presentation to the ciphertext and the key. `IdentityRegistry.register(issuer, pk, E, (A, B), proof)` verifies the proof via BN254 precompiles in 443,892 gas (measured). After this call `isVerified[alice] = true` and Alice can touch BUCK. She repeats steps 2-3 for every Ethereum account she controls; the issuer never sees those accounts.

10.2 Step 1: Mint

Alice calls `BUCK.mint(50_000e6)` – the no-arg form, so the built-in selector sorts her credits by `premiumRate` ascending: LandCo (50bp) first, then HomeCo (200bp). At `BUCK_K = 1e18`, and writing BUCK amounts rather than raw 6-decimal units:

1. `buckK.compute()` runs the PID cadence. The static controller returns `fundingFactor() = 0=`, so the counter-cyclical reserve gate is disabled and no pre-mint balance is captured.
2. The allocator walks #42 first: $e = 50bp \cdot 10 = 0.05$, $K - e = 0.95$, `avail = 200,000 - 0`, `netCap = avail * 0.95 = 190,000`. Since `netCap >= 50,000` the closed-form partial fill lands it in one credit: `take = ceil(50,000 / 0.95) = 52,631.578948`, `principal = take - 50,000 = 2,631.578948`. #43 is never touched. (At $K = 0.75$ the same mint would take $50,000 / 0.70 = 71,428.571429$ of #42 and deposit 3,571.428571: Alice still gets 50,000.)
3. `activateFromBuck(#42, alice, take)` grows `activatedValue` to `take`, and `mintsBacked[#42]` with it. From this moment #42 cannot be transferred.
4. `insurancePool` is credited the principal and Alice's signed raw is debited the same: the principal's 10 % assumed ROI (263.16/yr) exactly funds the 50bp annual premium on 52,631.58 of coverage (263.16/yr), in perpetuity.

End state, all of it live-derived rather than stored:

```
activatedValue[#42] = mintsBacked[#42] = 52,631.578948
creditLimit(alice)      = 52,631.578948    (#42 is NONE-schedule, so no haircut)
signedRawBalanceOf(alice) = -2,631.578948
balanceOf(alice)        = 50,000.000000    (limit - used)
totalSupply              += 2,631.578948    (the pool's positive contribution)
```

Alice now holds no BUCK at all and can spend 50,000 of it. Had the selector reached #43 instead, she would have got the same 50,000 – the gross-up in `mint(N)` and `burn(N)` would simply have activated more of #43's face to carry it, and charged premium on the 285,600 appraisal rather than the 300,000 the structure was worth when it was new.

10.3 Step 2: Approve Bob (Identity Handshake)

Alice's wallet reads Bob's `pk_bob` from `IdentityRegistry`, decrypts `E_alice` (it knows `sk_alice`) to recover `M`, and re-encrypts under `pk_bob` with fresh randomness as `E_bob = (r_b · G, r_b · pk_bob + M)`. It produces a Chaum-Pedersen NIZK `pi_CP = (e, s_1, s_2)` proving `E_bob` encrypts the same `M` registered in `E_alice`, with Fiat-Shamir bound to `msg.sender = alice`, `spender = bob`, `chainid`.

`BUCK.approve(bob, 10_000e6, E_bob, pi_CP)` verifies both parties are `isVerified`, runs `IdentityRegistry`. (29K gas), stores `~receiptFragments[alice][bob] = keccak256(E_bob)`, calls `markApproved(bob)` to freeze his `isCarrying` flag, and writes the ERC-20 allowance.

Off-chain (e.g. via Holochain), Alice delivers the matching `identity_data` and `E_bob` ciphertext to Bob’s wallet. Bob decrypts to recover `M'` and asserts $H(\text{identity_data}) * G == M'$ – the two-channel binding from the Identity document.

10.4 Step 3: Bob Pulls the Transfer

`BUCK.transferFrom(alice, bob, 10_000e6)` debits the allowance, runs `_identityCheckedTransfer` (both parties verified, `_receiptFragments[alice][bob] != 0`), and dispatches to the Non-Carrying path (Alice is not Carrying). Tokens move; `BuckTransferReceipt(alice, bob, 10_000e6, fromHash, toHash)` is emitted. An auditor with subpoena power over either Alice or Bob can recover the underlying identities; an outside observer learns only that addresses Alice and Bob exchanged 10,000e6.

10.5 Step 4: Alice Denominates as Notes

Alice picks flavour `A2` and values `[100e6, 25e6]` (totalFace 125e6). Her wallet pads the live pair with 14 dummies (`v=0`) so the batch lands on the smallest pinned circuit (`N=16`), snapshots (`oldRoot, nextLeafIndex`) from chain, derives the depth-20 sibling paths in submission order, and runs `snarkjs groth16 fullProve` against `build/snark/mint_batch_n16/mint.wasm` and `mint_final.zkey`. The proof triple is abi-encoded into `proofBytes`.

On-chain:

1. `BUCK.approve(notes, 125e6, E_notes, pi_CP_notes)` – the always-CP rule applies even though Notes is bound under a Public Identity. The receipt fragment is openable on subpoena.
2. `Notes.mint(proofBytes, oldRoot, newRoot, nextLeafIndex, 125e6, [cm[0]..cm[15]])` – the adapter dispatches internally by `cms.length=16` to its pinned `MintBatch16` Groth16 verifier. On success, 125e6 BUCK moves from Alice to Notes, `roots[]` advances to the SNARK-attested `newRoot`, `nextLeafIndex` advances by 16, `noteFaceSum + 125e6=`. Alice keeps the live openings privately; the dummies are discarded.

10.6 Step 5: Spend a Note

Later, the holder of one of Alice’s notes (Alice herself, or a bearer she off-chain handed the opening to) redeems `cm[0]` for 100 BUCK to recipient:

1. The wallet reconstructs (`pathElements[20], pathIndices[20]`) for `cm[0]` by replaying the `Minted` event log and pulling `cms[]` from each batch’s calldata.
2. `snarkjs groth16 fullProve` against `spend.wasm / spend_final.zkey` produces a proof binding `[currentRoot, Poseidon-3(T_NF, rho, idHash), 100e6, recipient, 1]` to the witness.
3. `Notes.spend(proofBytes, currentRoot, nullifier, 100e6, recipient)` – the contract checks the root window, rejects a previously-seen nullifier, runs the adapter (~360K gas total), marks the nullifier burned, decrements `noteFaceSum`, and calls `buck.transfer(recipient, 100e6)`. Notes is bound Carrying, so the transfer dispatches to the carrying path automatically: the recipient receives the full 100e6 and absorbs an amount-weighted share of the pool’s average demurrage age via `buckSeconds`.

If HomeCo later reappraises the structure at \$280K, that is allowed – #43 carries no activated coverage, so the floor in Insurer Updates does not bind, and Alice’s future capacity against it drops accordingly. What HomeCo *cannot* do is write #42’s coverage down: LandCo is #42’s insurer, and even LandCo may not reappraise below the 52,631.58 Alice has bought.

Meanwhile #42 keeps depreciating on its schedule – NONE, so not at all here, but a depreciating credit would pull `creditLimit` down under Alice continuously. If it fell below her drawn 2,631.58 she would be underwater: she could not burn until she reacquired BUCK and repaid, her insurance would stay in force regardless, and the Jubilee relief accruing against her lien would shrink the cost of closing the position year on year.

11 Gas Cost Snapshot

Approximate gas costs from the current Foundry suite (Solidity 0.8.28, cancun, via-IR, optimizer at 200 runs). PID and oracle figures are estimates against unwritten oracle wiring.

Operation	Gas (approx)	At 30 gwei	Notes
BUCK_CREDIT.updateCredit()	~80K	~\$6	Multiple SSTORES (in
IdentityRegistry.register()	~444K	~\$36	credential proof, 3 pair
IdentityRegistry.verifyApprove	~29K	~\$2	CP proof, per counterp
BUCK.approve() (identity)	~97K	~\$8	29K CP + 46K allowa
BUCK.transfer / transferFrom	~80K	~\$6	Identity check + receip
– reentrancy guard, per guarded call	~600	–	TSTORE/TLOAD; no
– credit scan, sender holding 10 credits	~28K	~\$2	Live creditLimit; 6K w
– demurrage routed to a designated payer	~900	–	Warm payer slot; 0 if n
BUCK.mint() (static BUCK_K, 1 NFT)	~150K	~\$12	NFT walk + allocator
BUCK.mint() (static BUCK_K, 2-NFT spillover)	~210K	~\$17	+ second NFT crystal
BUCK.burn() (1 NFT)	~120K	~\$10	Mirror of mint
BUCK.mint() (with PID update, 4-pool basket)	~450K	~\$36	+ 4 V3 TWAP reads -
BUCK.mint() (PID cached, within dT)	~155K	~\$12	one SLOAD on the ca
Notes.mint (public issuer, N=1)	~513K	~\$41	Groth16 verify + batch
Notes.mint (private issuer A2, N=1)	~629K	~\$50	+ re-encryption bindin
Notes.spendCoupledA1	~964K	~\$77	spend proof + folded g
Notes.spendCoupledA2	~982K	~\$79	spend proof + folded g
Notes.spendCoupledB1	~906K	~\$72	+ depositor sigma + n
BuckCredit.attestInsurer (8 predicates)	~8M	~\$640	once per review period
BUCK_CREDIT.currentValue / BUCK_K.currentBuckK	0	Free	View

The PID update cost is amortised across all mints within a dT window (60 s default; only the first mint after each window pays the oracle-read + PID-math cost; every other mint sees the cached one-SLOAD path). `Buck.burn` calls `compute()` too, so burns share the amortisation. The same contracts deploy on EVM L2s (Arbitrum, Optimism, Base) where gas is 10-100x cheaper.

12 Security Considerations

12.1 Oracle Manipulation

BUCK/USDT and basket pool prices use Uniswap V3 TWAPs (600 s shipped, governance-tunable per pool), so manipulation requires sustained capital commitment over the full window proportional

to pool depth. The `BuckKControllerV3TwapTest` suite verifies that a single-block flash-shape attack driving BUCK to \$0.50 leaves spot collapsed but TWAP unchanged at $\sim \$1.000002$, with the PID computing off the unchanged TWAP. When/if Chainlink basket feeds are adopted later, the Chainlink branch of `_getBasketCost` must verify `updatedAt` is within an acceptable window (~ 1 hour) and revert if stale.

12.2 The Insurer Gate

A credit's `faceValue` and `premiumRate` are whatever its issuer says they are, and a zero-premium credit costs nothing to draw on – `poolPrincipal == 0`, so the counter-cyclical funding gate is inapplicable by design. If becoming an insurer were ungated, a single verified account could write itself a policy on an asset it never names, and mint against it:

```
mallory issues herself a zero-premium credit, faceValue 1,000,000,000
mint(500,000,000)    -> principal paid    0
                        spendable          500,000,000 BUCK
```

The recipient opt-in below cannot stop this, since the attacker is both insurer and client. What stops it is the regulator. Insurer oversight is already a periodic review of underwriting, reserves and reinsurance that concludes with an estimate of what the insurer may write; the gate makes that conclusion available to the contract, as a set of public memberships the insurer proves and `BuckCredit` caches.

The envelope. An insurance regulator keeps one public subtree per predicate in the identity accumulator, named under its namespace (e.g. `regulator:ca-ab`): standing (`insurer`), a face band on a ladder of powers of ten from ten BUCK to a hundred million (`insurer:face:<b>`), each permitted depreciation model (`insurer:dep:<d>`), the maximum depreciation and premium rates in basis points (`insurer:depRate:<bp>`, `insurer:premium:<bp>`), the general scope (`insurer:general`), and each asset scope it permits (`scope:<asset path>`). Subtree identifiers and scope identifiers are `keccak` of the namespaced name, so refining a scope needs no version and there is no asset taxonomy on chain. An insurer's leaf is the public, unsalted identity leaf: its licence is a fact it advertises.

Attestation, once per review period. `attestInsurer(M, opening, claim, root, paths)` takes the insurer's identity point, a Chaum-Pedersen opening showing the caller's registered credential decrypts to `M` (bound to the account, chain and registry under its own `v2` tag, so no account can claim another's envelope), the claimed envelope, a posted root the registry accepts for the insurer consumer (one day old at most), and one public path per claimed predicate. It verifies each path with `verifyPublicMembership` and caches the envelope with `expiresAt = rootPostedAt(root) + attestationPeriod`. A new attestation withdraws every scope it does not repeat. Eight predicates cost about 8M gas, once per period.

Issuance, every credit. `createCredit` keeps its eight-argument form, which declares the general scope, and gains a nine-argument form declaring a `scope`. Either checks the credit against the cached envelope with a handful of storage reads, reverting with the reason: "insurer not in good standing", "attestation expired", "scope not attested", "face above attested band", "depreciation model not attested", "depreciation rate above attested maximum", "premium rate above attested maximum". `updateCredit` is held to the same parameter envelope, or an insurer would write small and reappraise large; the scope is emitted at creation and not stored, so a reappraisal cannot re-check it. `bandForFace`, `scopeId` and `scopeAttested` expose the arithmetic.

Configuration. The deployer calls `configureInsurerGate(registry, namespace, period)` once; it cannot be changed or removed. A `BuckCredit` never configured issues ungated – the

economic simulations deploy that way – and a deployment profile **must** configure it. The gate’s own suites (`InsurerGate.t.sol`, and `InsurerGateVectors.t.sol`, which replays the Python reference’s subtrees, opening and issuance decisions byte for byte) exercise every refusal.

The gate checks the consistency of the attestation chain, not its accuracy: a regulator that attests a bad insurer is answerable for it, off chain, exactly as it is today. Out of scope by decision: what happens to an insurer’s pools when it fails, which is its reinsurers’ business.

12.3 Recipient Opt-In

A credit only lands where its recipient asked for it. A client calls `setCreditIssuer(insurer, true)` before that insurer may issue to them; the flag is per-pair, revocable, and governs only new issuance, since credits already held may be backing BUCK. The rule is uniform – naming yourself the insurer does not exempt you – which keeps the invariant storable in one line and declines to open a carve-out shaped exactly like the attack above.

What this closes is the gas-griefing vector. A credit costs its holder gas forever after: `totalCurrentValue` walks `tokenOfOwnerByIndex` over *every* token the account owns, and since the credit limit is read live, that walk runs on every outbound transfer from a non-Carrying account. Without the opt-in an attacker could raise a chosen address’s transfer cost without bound for the price of the mints, and the victim could not refuse delivery – ERC-721 `_mint` needs no consent from the receiver. A holder can still dispose of an unwanted credit by transferring it away, since an unactivated credit is freely transferable, but that is a remedy they should never have had to reach for.

It is not a substitute for vetting. It bounds who may put a credit in your hands; the gate above bounds what they may write.

12.4 Insurer Trust

An insurer may reappraise `faceValue` downward, but not below coverage the holder has already bought (see Insurer Updates). So an insurer cannot cancel a policy, cannot strand a position, and cannot by itself put a holder into default: the worst it can do is reduce future borrowing capacity against that asset, and – via the depreciation schedule – let the holder’s credit limit decline over time. A holder pushed below their limit that way is in the supported underwater state, not a defaulted one.

That leaves the residual risks from an insurer who *has* been vetted: refusing to honour a claim, and setting a punitive `premiumRate` on new coverage. Multi-insurer market discipline is the natural check, and a rating that responds to claim behaviour is the intended mechanism. Time-locks on insurer updates (e.g. 7-day delay for reductions > 20 %) remain a future enhancement. For the prior question – how an insurer becomes vetted at all – see The Insurer Gate.

12.5 Reentrancy

Every state-mutating entry point on `Buck` carries a transient-storage mutex; the design and the measured cost are in Reentrancy Guard. The guard has no exemptions.

The guard is defence in depth rather than a fix for a live hole. Every outbound call `Buck` makes today is either a `STATICCALL` – including the PID controller’s oracle reads, which sit behind an `internal view _readReferences()` – or a call into one of the three immutable, address-locked system contracts, none of which reach arbitrary code. What the guard protects against is the future: `BuckCredit` behind a UUPS proxy (see Future Directions), or any ERC-721 receiver hook added to the activation path. The window it closes is real and specific: `_allocateMint` activates coverage

– which grows credit headroom immediately – before debiting the minter the pool principal, so `mid-mint balanceOf` overstates, and `_subBalance` does not re-check the limit on the way out.

Note that view functions are not guarded and cannot be: read-only reentrancy against `balanceOf` is a hazard for *integrations*, which must not sample a BUCK balance from inside a callback they do not control. See Integrating with BUCK.

12.6 The Insurance Pool: an Interim Stand-in

Today. `insurancePool` is one address, fixed at construction, that receives every mint’s premium deposit, pays every burn’s refund, and is the authority that wires the basket (`setBasket`). It holds the cumulative pool principal for every outstanding mint. Burn requires the pool’s spendable balance to cover the proportional refund; if the pool is drained out-of-band the burn reverts. Every test and simulation constructs the pool as a contract bound Carrying through the registry – the Foundry suites with `bindCarryingPool`, the Python and JavaScript worlds with a `SimLP` their operator binds – so its deposits keep their age, and `burn` dispatches the refund on the registry’s flag like any transfer: a Carrying pool hands the refunded share of its accrued age back to the holder. (The Notes paper’s fixture world, `notes_stack.py`, leaves its pool unbound: its credits carry no premium, so no deposit ever reaches it, and binding would disturb its pinned fixtures.) A bound pool is a verified identity, so a transfer out of it is possible, and the simulations’ `SimLP` lets anyone call its `exec`: a stand-in, not a vault. It stands in for the design below until insurers exist on chain.

Intended (author, 2026-09-25). A credit’s premium deposit belongs with that credit’s insurer – `credits[tokenId].insurer`, the account that wrote it. The insurer holds its credits’ deposits in a premium pool: a hot wallet that invests the deposits to earn the premiums, and that it must keep funded to cover every upcoming cancellation (the refunds due when holders drop their insurance) and every upcoming claim payout. The pool works on its members’ behalf, so it is Carrying – a contract bound through the identity registry with `isCarrying`, like any other account doing work for others, not a special case in Buck (an EOA cannot be Carrying). Buck then takes no pool address at all:

- `mint` credits each slice’s pool principal to that slice’s insurer, and `burn` refunds each slice from its insurer; the funding gate is unchanged (it sees the total).
- Solvency becomes per insurer: `pool(insurer) >= sum of mintsPrincipal[tokenId]` over the insurer’s credits, plus what its outstanding claims require. A burn against an underfunded insurer reverts, so an insurer that lets its hot wallet run dry traps its holders’ cancellations – a condition for the insurer gate’s attestation to police.
- The basket’s wiring moves to the registry’s governance (`identity.governance()`).

What the change touches, when the insurer implementations – the insurance company’s interface to the premium pool its credits fund – are built:

- Buck: drop the constructor parameter; route mint principals and burn refunds per slice; move the `setBasket` authority; per-insurer `Transfer` events.
- `BuckCredit` and `BuckTypes`: `CreditSlice` gains the insurer, which `batchCreditInfo` returns.
- Deployments: `scripts/deploy/Deploy.s.sol`, `alberta_buck/sim/deploy.py` and `notes_stack.py`, `core/js/src/buckworld.js` with the equilibrium world’s `setBasket` and the sandbox.

- Tests: the sixteen Foundry suites that construct `Buck`, `Buck.t.sol`'s pool and `setBasket` cases and the basket suites' wiring; the JavaScript market and sandbox tests that read the pool.
- The published contract bundle: `Buck`'s constructor and `batchCreditInfo` change, a new minor version.

13 Future Directions

This implementation prioritises clarity and correctness; alternatives the architecture document flags for later study:

- **ERC-1155 batch credits** if standardised credit *classes* emerge (e.g. "Calgary residential, HomeCo, LINEAR 200bp"), enabling `safeBatchTransferFrom` for portfolio management and reduced gas for accounts holding many same-class credits. Per-token depreciation start dates partially negate the fungibility benefit, so applicability depends on the secondary market.
- **ERC-3643 (T-REX)** if regulatory KYC compliance becomes a deployment requirement. Native ONCHAINID, programmable transfer restrictions, agent-role granularity, and lost-key recovery via identity verification. This is the likely production standard if Alberta deploys BUCK under provincial regulation.
- **ERC-4626 premium pools** – a vault shape for each insurer's premium pool (see "The Insurance Pool: an Interim Stand-in"), so deposits become standardised shares with `deposit` / `shares` / yield composability.
- **L2 deployment.** The contracts are EVM-compatible and deploy on Arbitrum, Optimism, Base, or Polygon with no changes. Cross-chain BUCK via canonical bridges or LayerZero/Axelar would enable multi-chain liquidity while keeping a single BUCK_K oracle on the settlement layer.
- **Coverage-ratio premium.** The current per-NFT `premiumRate` is flat. A loss-layer-style effective-rate derivation – $\text{rate}(V, A, D) = \text{lambda} * \text{coverage} * (1 - D/V)^2 / 2$ under uniform loss – would price closer-to-full-coverage activations more aggressively, which matches actuarial intuition for parametric insurance with deductibles. The allocator's sort key would move from raw `premiumRate` to a midpoint-evaluated effective rate; everything else carries through.
- **Recursive SNARKs above N=512.** Mint batches above N=64 hit the EIP-170 24 KB bytecode ceiling on stock Groth16 verifiers. Calldata-IC delivery (hash-pinned) buys headroom to N=128 or 256; Nova / Halo2 / Plonky3 recursion is the natural fix above N=512.
- **Buck as a Diamond (EIP-2535).** Buck today is a single monolith covering identity, demurrage, NFT-credit accounting, ERC-20 wire-protocol, BuckBasket direct-mint hooks, and Jubilee bookkeeping. Splitting these into facets (identity facet, demurrage facet, mint-burn facet, ERC-20 facet) is the natural modularization path: facets share Buck's storage layout (`AccountState`, `_totalSupply`, `mintsBacked`), function selectors don't collide within a single ERC standard, and per-facet upgradability lets bug fixes ship without redeploying the whole stack. Cross-contract calls into `BuckCredit` and `BuckKController` stay external – the ERC-20 / ERC-721 selector collision detailed in BUCK_CREDIT forces `BuckCredit` to remain a sibling.

- **Independently upgradeable BuckCredit (UUPS or transparent proxy).** Keep Buck's `immutable buckCredit` address stable while patching BuckCredit's logic in place. Combined with the Diamond split above, this lets the system evolve facet-by-facet and contract-by-contract without the Big Bang redeploy that an immutable monolith requires.

14 Status

Current Foundry suite: 55 test files cover all six contracts in tree (`BuckCredit`, `IdentityRegistry`, `Buck`, `BuckBasket`, `BuckKControllerStatic` + `BuckKControllerDirect` + `BuckKController`, `Notes`) plus dedicated suites for the NFT-credit machinery (`BuckCreditLimit.t.sol`, `BuckCreditBacking.t.sol`, `BuckCreditReappraisal.t.sol`, `BuckSignedBalance.t.sol`), the reentrancy guard and delegated demurrage (`BuckReentrancy.t.sol`, `BuckDelegatedDemurrage.t.sol`, `GuardBench.t.sol`), the identity accumulator and insurer gate (`IdentityAccumulator.t.sol`, `InsurerGate.t.sol`, `InsurerGateVectors.t.sol`), the SNARK toolchain (`mint_batch[_a2].circom` at pinned $N \in \{1, 2, 4, 8, 16, 32\}$, `spend.circom`, the two folded deposit gates and BI's membership, each with real proofs), and a Python reference wallet (`alberta_buck.wallet`) that emits canonical test vectors the Solidity tests load via `vm.readFile`. One `BuckKControllerForkTest` requires a live mainnet RPC and is skipped offline.

The `BUCK_K` and `BuckBasket` test suites cover:

- `BuckKControllerBaseTest` – abstract-base harness exercising the shared PID mechanics (priming, `dS` compensation, `dT`, `dTMax`, anti-windup, integral convergence, `fundingFactor`) against a synthetic `_readReferences` override.
- `BuckKControllerStaticTest` – governance-set constant and `setBuckK` guards.
- `BuckKControllerDirectTest` – the USD-free direct embodiment wired to a mock basket source; `reprime` access control, dilution-discontinuity absorption, sign-convention canaries on the `basketValue > 1.0` and `< 1.0` branches.
- `BuckKControllerUnitTest` – harness with mocked Chainlink feeds, mocked BUCK price. Includes priming, `dS` compensation, `dTMax` clamp, anti-windup, and multi-step integral convergence.
- `BuckKControllerV3Test` – locally deployed Uniswap V3 factory, all four basket pools at reference prices, BUCK/USDT pool with a $1e17$ full-range LP (~ 100 K BUCK / 100 K USDT, matching the design narrative). Spot-price reads, swap-driven price drift, and PID response.
- `BuckKControllerV3TwapTest` – same V3 setup with `twapInterval = 600 s` on every pool, plus a warmup loop that walks forward writing observations on each pool. Verifies the TWAP path smooths a flash-shape spike (spot \$0.50, TWAP \$1.000002) and tracks sustained drift over the 600-s window (TWAP converges from $\$1.00 \rightarrow \$0.97 \rightarrow \$0.95$ over a full window at \$0.95 spot).
- `BuckBasketTest` – direct-mint deposit/redeem flows, `addBasketToken` renormalization, proportional-allocation correctness across overweight/underweight pools, "sell high / buy low" redemption invariants, shortfall-cover greedy swap, and the `reprime` callback into `BuckKControllerDirect`.

The Python simulation harness in `alberta_buck/sim/` externally drives an anvil instance with the deployed contracts (see `make sim*` targets and the project memo on the externally-driven sim) to replay multi-agent scenarios – `direct_mint.py` and `scenario.py` for the basket-side flows,

`agents.py` for arb / depositor / redeemer agents, `rebalancer.py` for the cross-pool recycling sanity check, `plot_basket_model.py` / `plot_rebalancing.py` / `plot_routing.py` for visualization.

Outstanding work tracked in companion memos:

- **Equilibrium-seeking dynamic-issuance test** – closed-loop integration of `BuckBasket`, `BuckKControllerDirect`, `BuckCredit`, and `Buck` against an external market driver and an arb agent that responds to `buckK` changes by minting/burning `BuckCredit` backed BUCK. Should emit a snapshot tape suitable for the existing `plot_*` visualizers and double as the regression for Dynamic Issuance: Why `BuckBasket` Needs BUCK_K. Highest-priority validation for the closed-loop control claim.
- **Resolve BUG #5..#11/#13 in `BuckBasket`** – per-withdrawal slippage guards, TWAP-bounded swap limits, TWAP-driven overweight/underweight reads, proportional reinvestment, symmetric TOKEN-deposit routing, basket-mint accounting in `fundingFactor`, treasury-reinvest event, graceful tail-redemption.
- **Mainnet-fork BUCK_K test** – exercise `BuckKController` against the four real on-chain pools at a pinned recent block, verifying basket cost reads sensibly against live data. Cheap validation step before mainnet.
- **Multi-block historical replay** – a backtest harness that walks `BuckKController` through 12 months of EVM history at a chosen block cadence, for tuning `Kp/Ki/Kd` against a real price tape. Highest-value validation; biggest engineering lift.
- **Serialized B1 sub-notes** (`spend_serialized.circom` + per-parent bitmap) – design validated end-to-end by `scripts/snark/serialized_notes_poc.py` at `N_fam` in `{256, 1024, 4096}`; circuit port and contract bitmap support are the implementation gap. See alberta-buck-notes-serialized.org.
- **Production trusted-setup ceremony** – replace dev entropy in `scripts/snark/setup.sh` with a multi-party ceremony per pinned circuit before mainnet.

The cryptographic surface is described by an executable Python reference; CI regenerates the canonical JSON test vectors and `git diff --exit-code`'s them against committed copies, so any drift between the spec, the wallet, and the Solidity tells us immediately which of the three is wrong. No `vm.ffi` shells out to Python at test time – the wallet emits vectors once, the Solidity reads them many times.